Multimodal Machine Learning

Lecture 5.2: Structured Representations and Reasoning

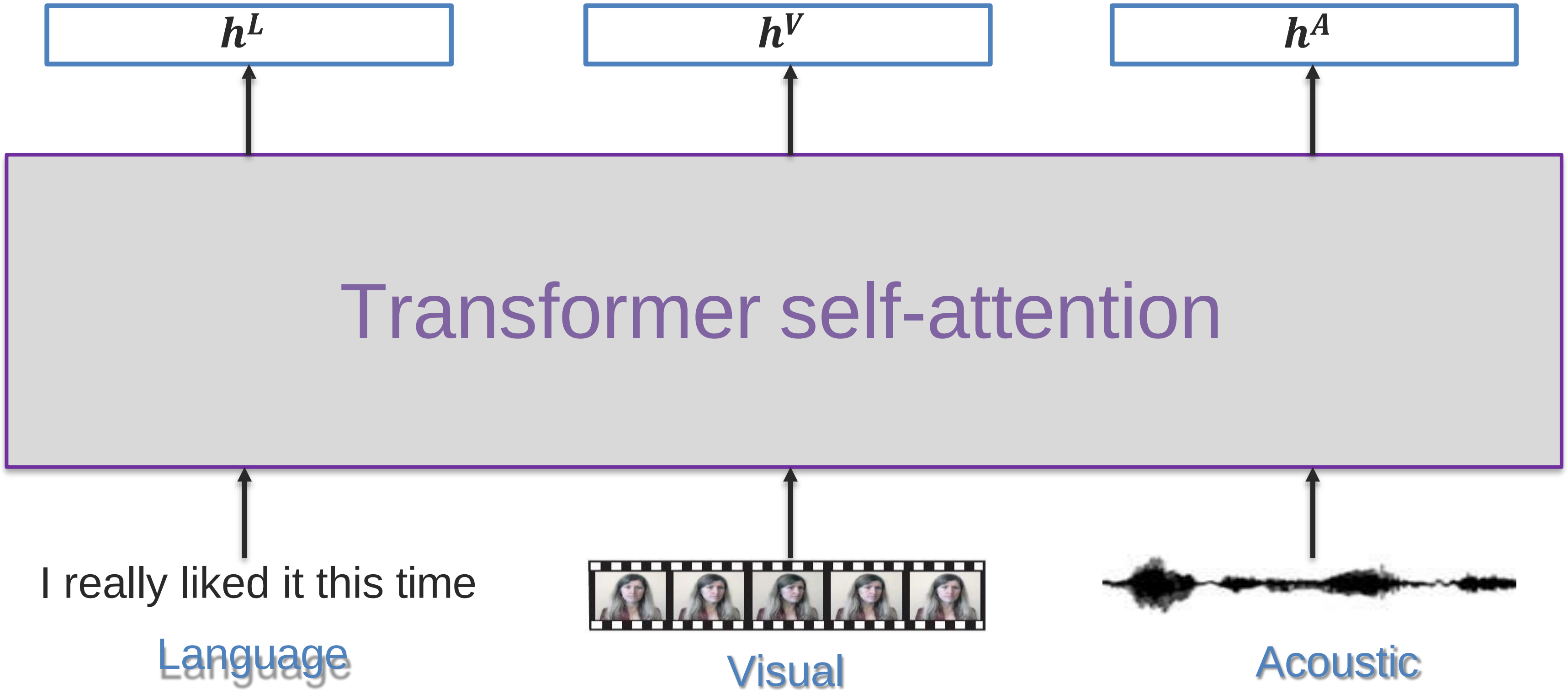
Dam Quang Tuan

Objectives of today's class

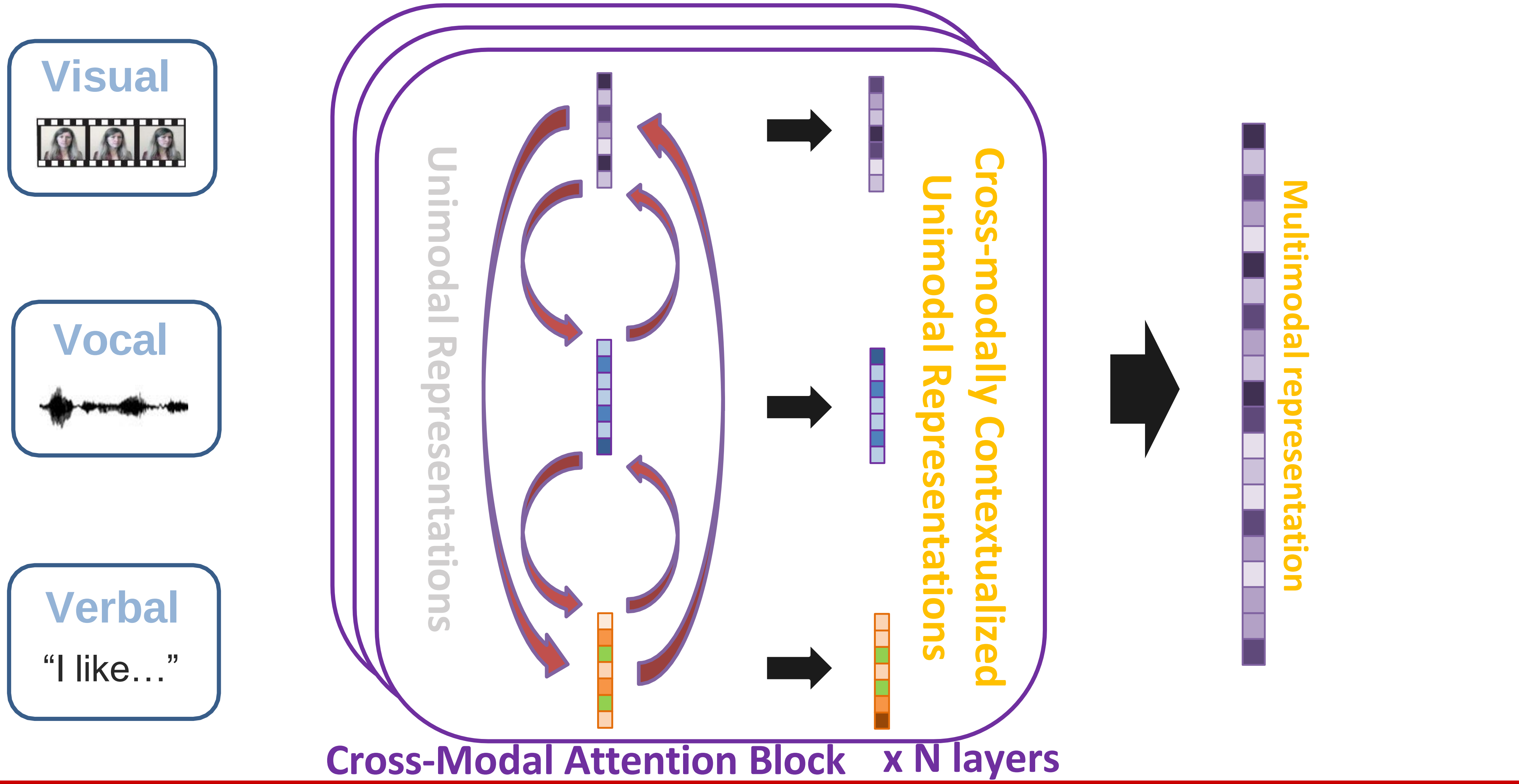
- Multimodal transformers
    - Modality-shift transformer (MAG-BERT)
- Sequence-to-sequence modeling with Transformers
- Going beyond sequences
    - Graph representations
        - Graph neural networks
    - Hierarchical representations
    - Modular representations
        - Neural module networks
        - Neuro-symbolic networks

Language-Vision Transformers

Simple Solution: Contextualized Multimodal Embeddings



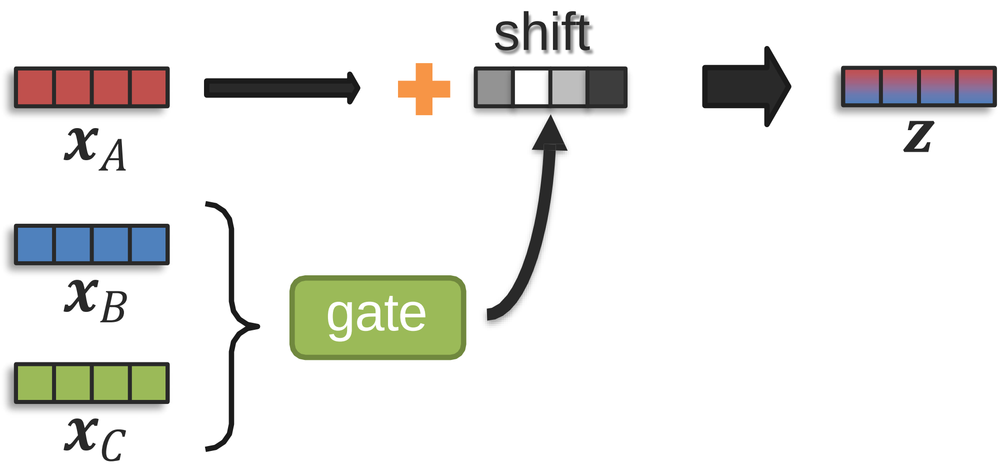
Multimodal Transformer – Pairwise Cross-Modal



Reminder: Modality-Shifting Fusion

Primary modality

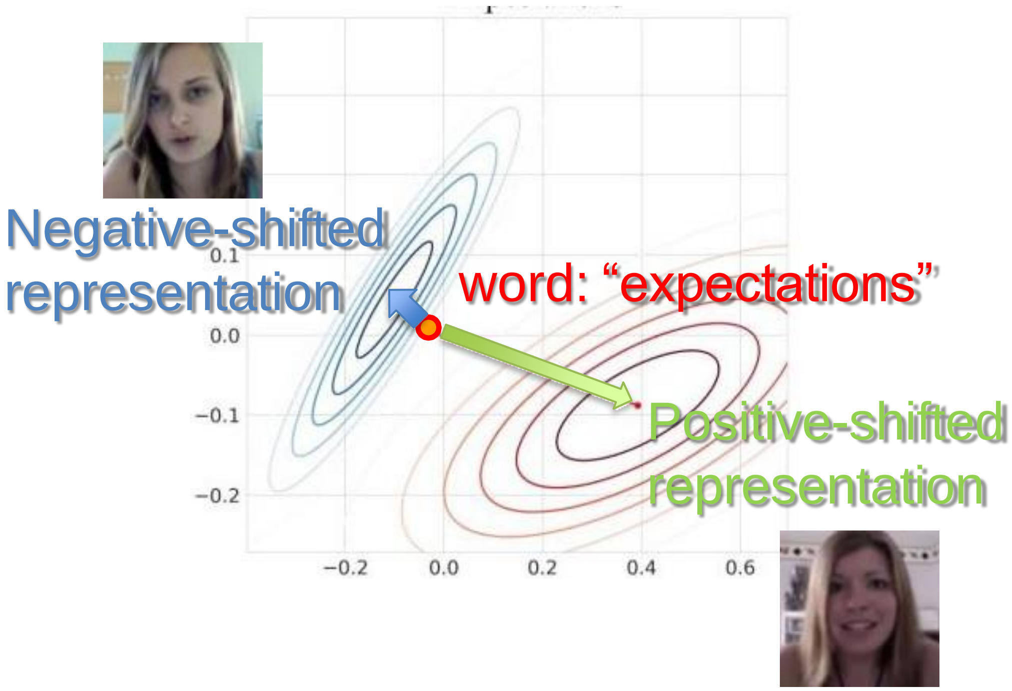
Secondary modalities



Example with language modality:

Primary modality: language

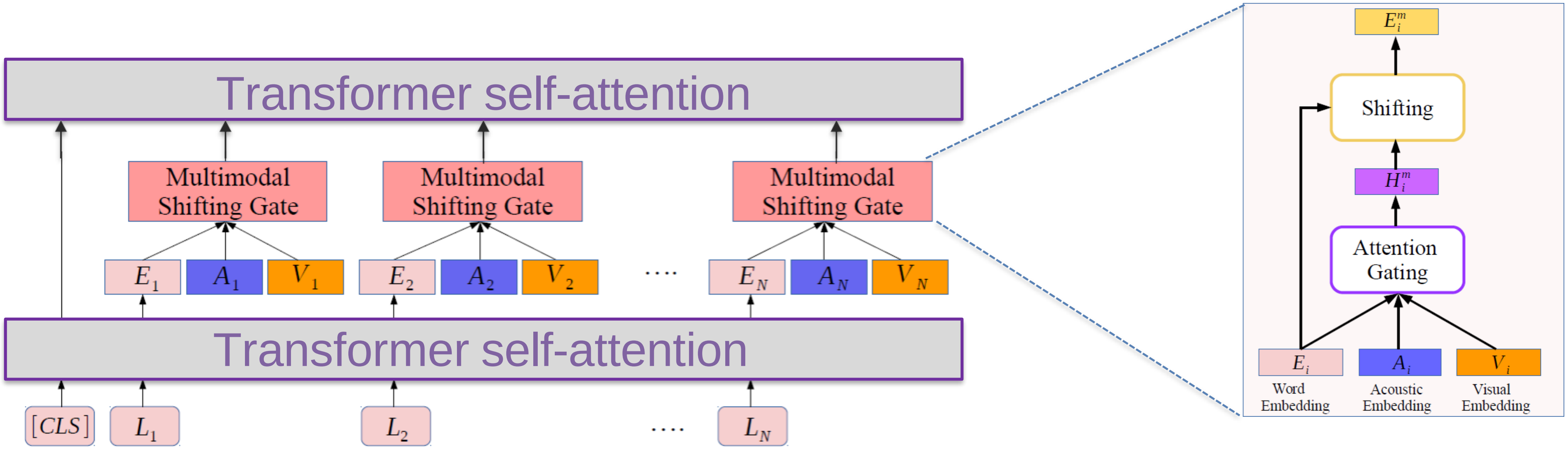
Secondary modalities: acoustic and visual



Wang et al., Words Can Shift: Dynamically Adjusting Word Representations Using Nonverbal Behaviors, AAAI 2019

Modality-Shifting with Transformers

Multimodal Adaptation Gate (MAG) + BERT

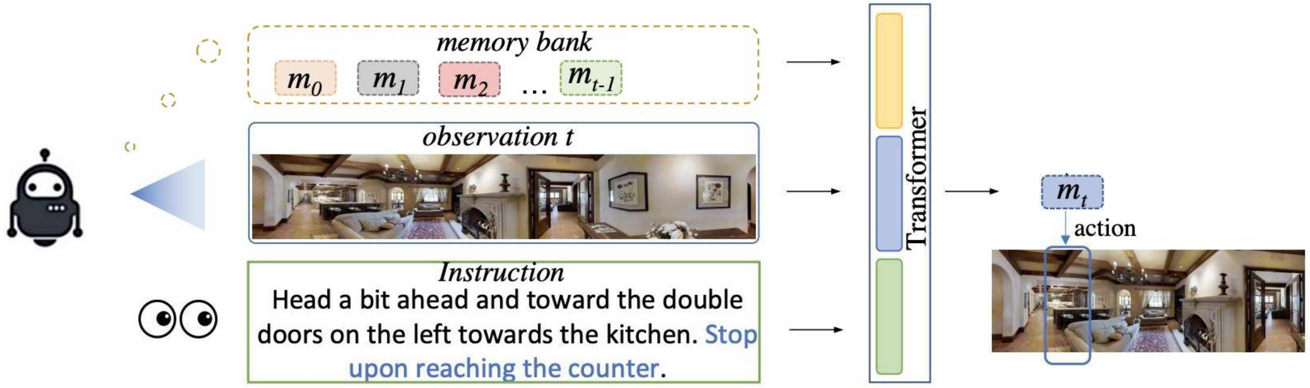


Rahman et al., Integrating Multimodal Information in Large Pretrained Transformers, ACL 2020

Memory for Multimodal Sequences

Memory + aligned contextualized representations

Where have I visited previously?



[Chen et al., History Aware Multimodal Transformer for Vision-and-Language Navigation. NeurIPS 2021]
[Lin et al., Multimodal Transformer with Variable-length Memory for Vision-and-Language Navigation. ECCV 2022]

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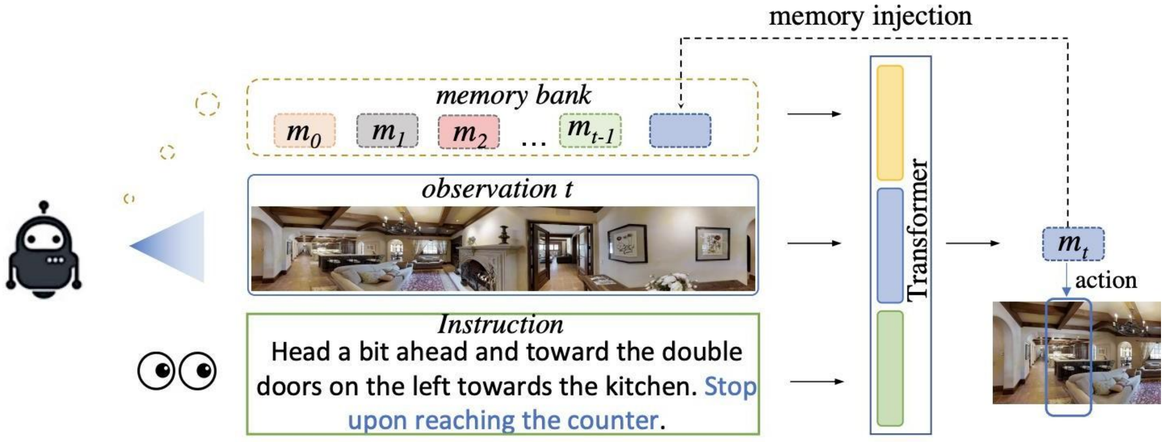

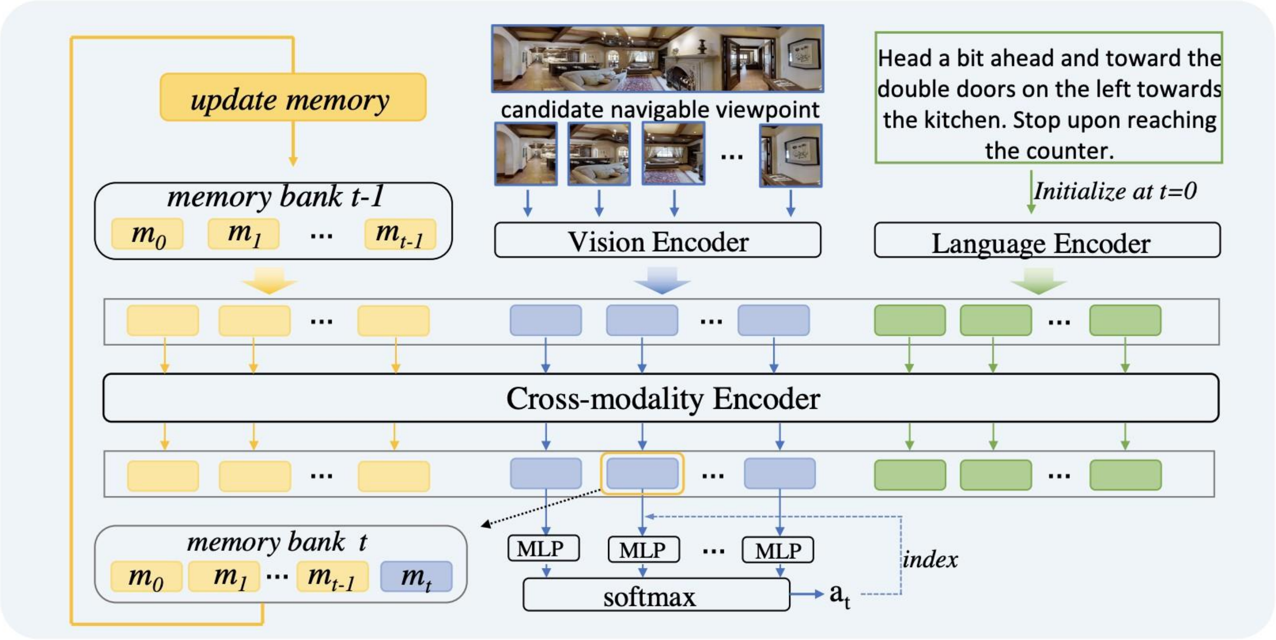

+ Contextualized representations

+ Memory mechanisms

[Chen et al., History Aware Multimodal Transformer for Vision-and-Language Navigation. NeurIPS 2021]
[Lin et al., Multimodal Transformer with Variable-length Memory for Vision-and-Language Navigation. ECCV 2022]

Memory for Multimodal Sequences

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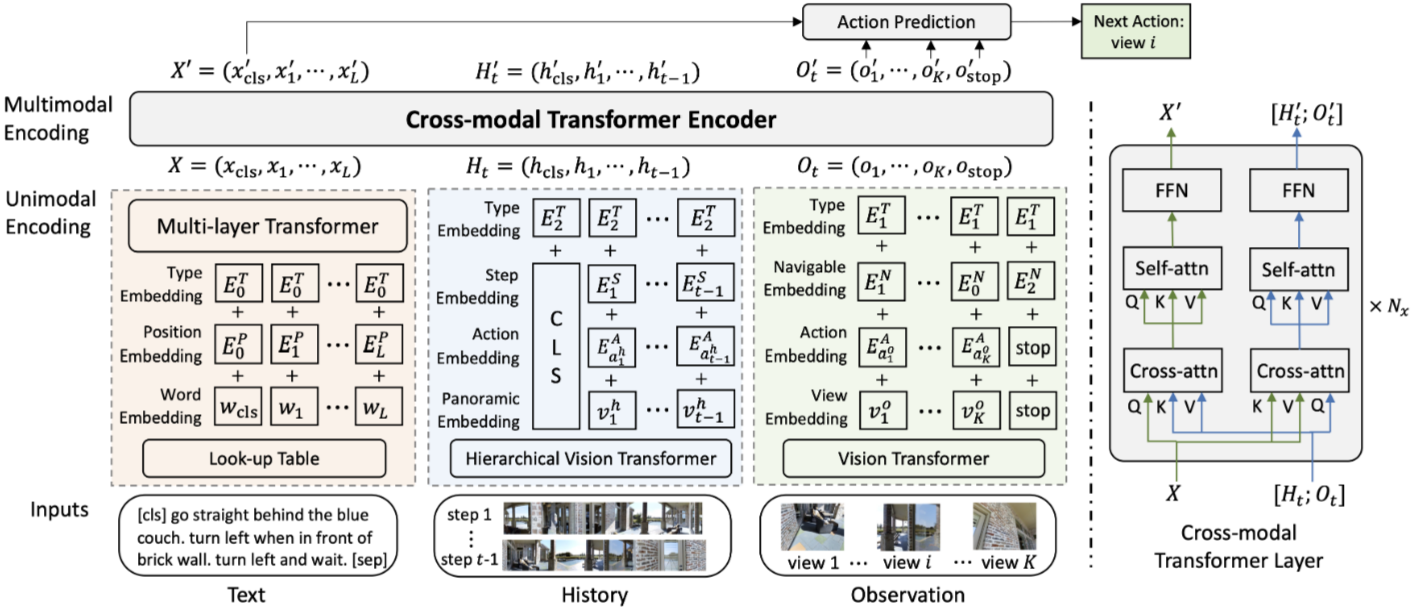

Figure 1: The architecture of History Aware Multimodal Tranformer (HAMT). HAMT jointly encodes textual instruction, full history of previous observations and actions, and current observation to predict the next action.

[Chen et al., History Aware Multimodal Transformer for Vision-and-Language Navigation. NeurIPS 2021]

Memory for Multimodal Sequences

Memory + aligned contextualized representations

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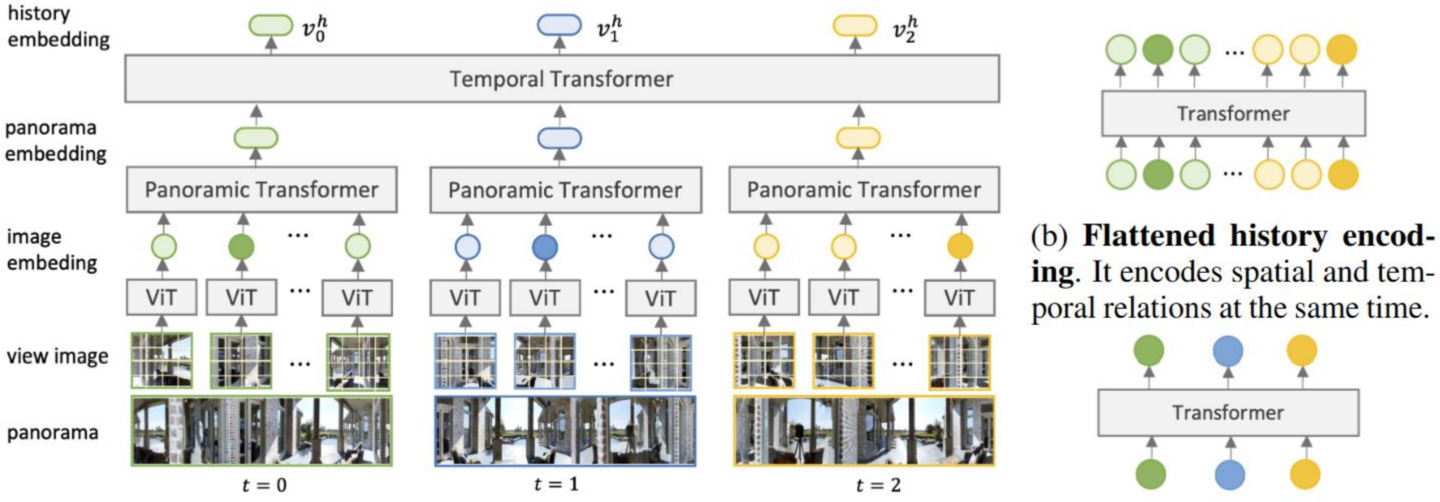


(a) **Hierarchical history encoding**. It first encodes individual view images with ViT, then models the spatial relation between images in each panorama, and finally captures the temporal relation between panoramas in the history.

(b) **Flattened history encoding**. It encodes spatial and temporal relations at the same time.

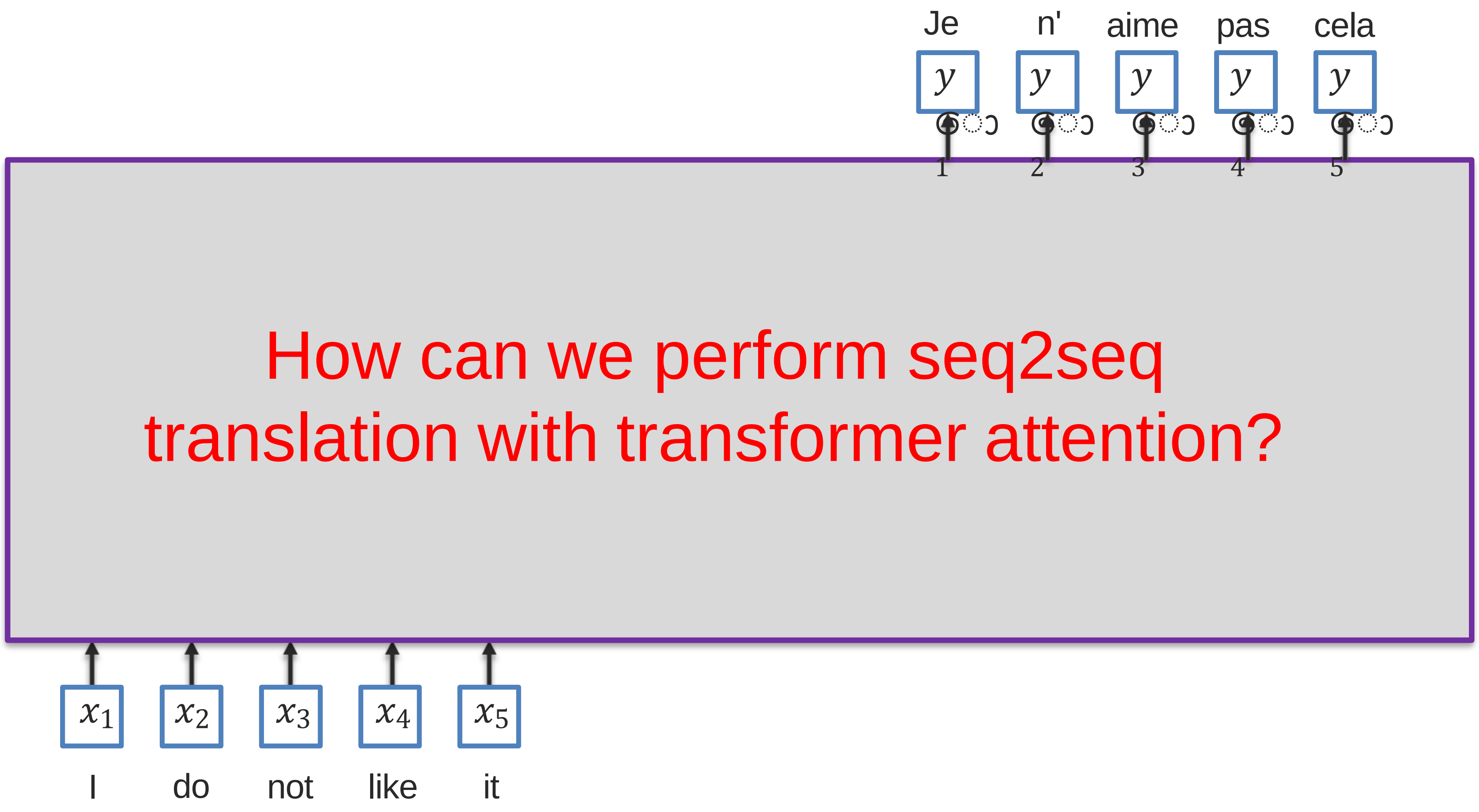
(c) **Temporal-only history encoding**. It only considers temporal relation of oriented views.

Figure 2: A comparison of history encoding methods. Circle nodes in different colors denote view images of panorama at different steps. Darker circle nodes are the oriented view of the agent.

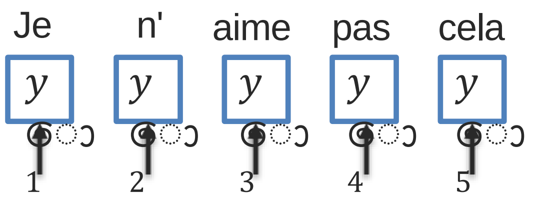
[Chen et al., History Aware Multimodal Transformer for Vision-and-Language Navigation. NeurIPS 2021]

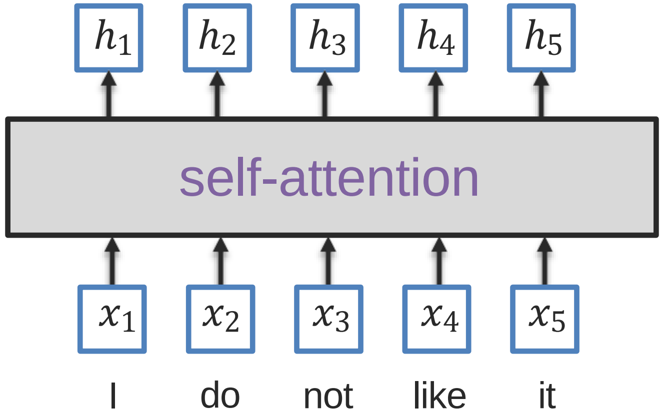
Sequence-to-Sequence Using Transformer

Sequence-to-Sequence Modeling

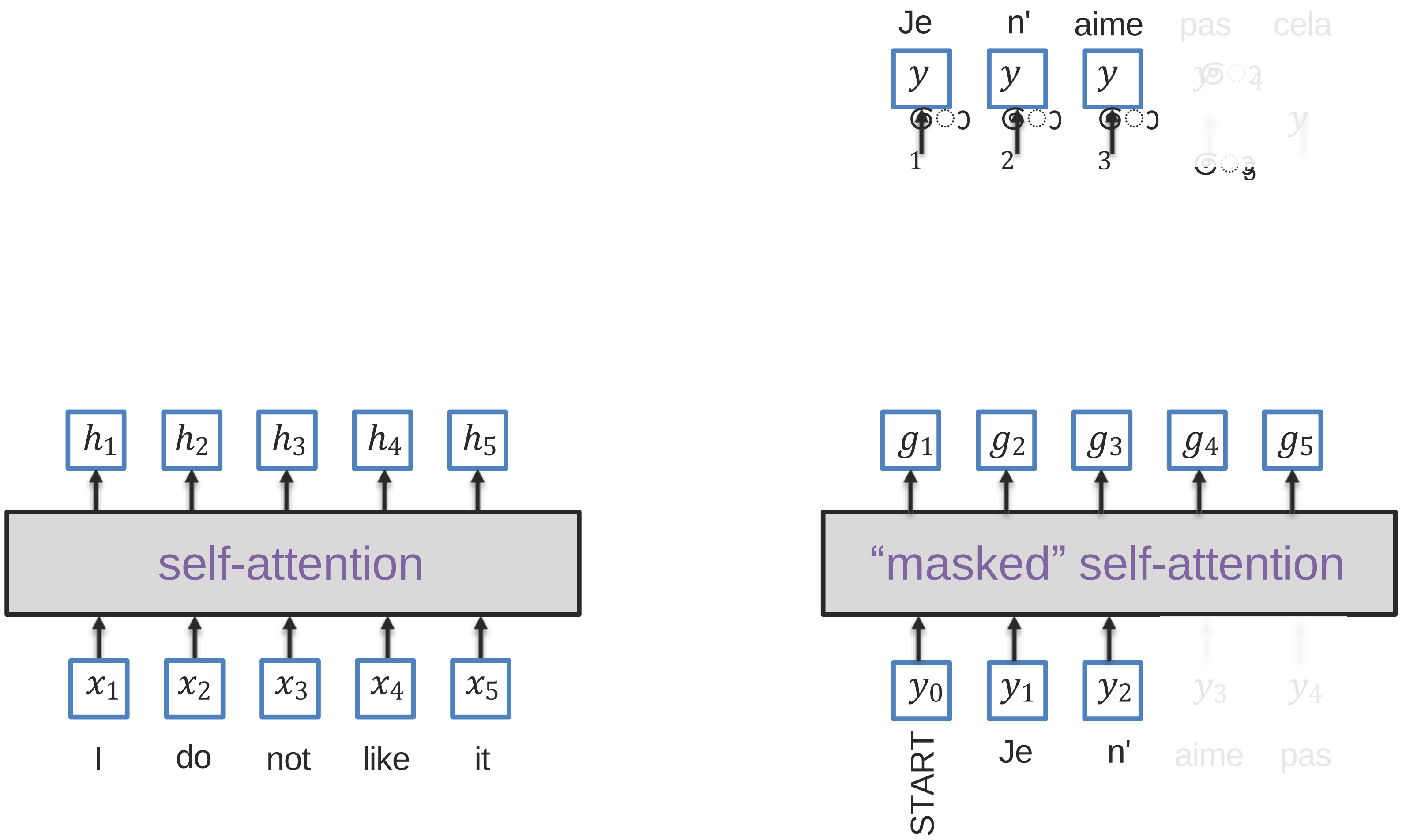


Seq2Seq with Transformer Attentions



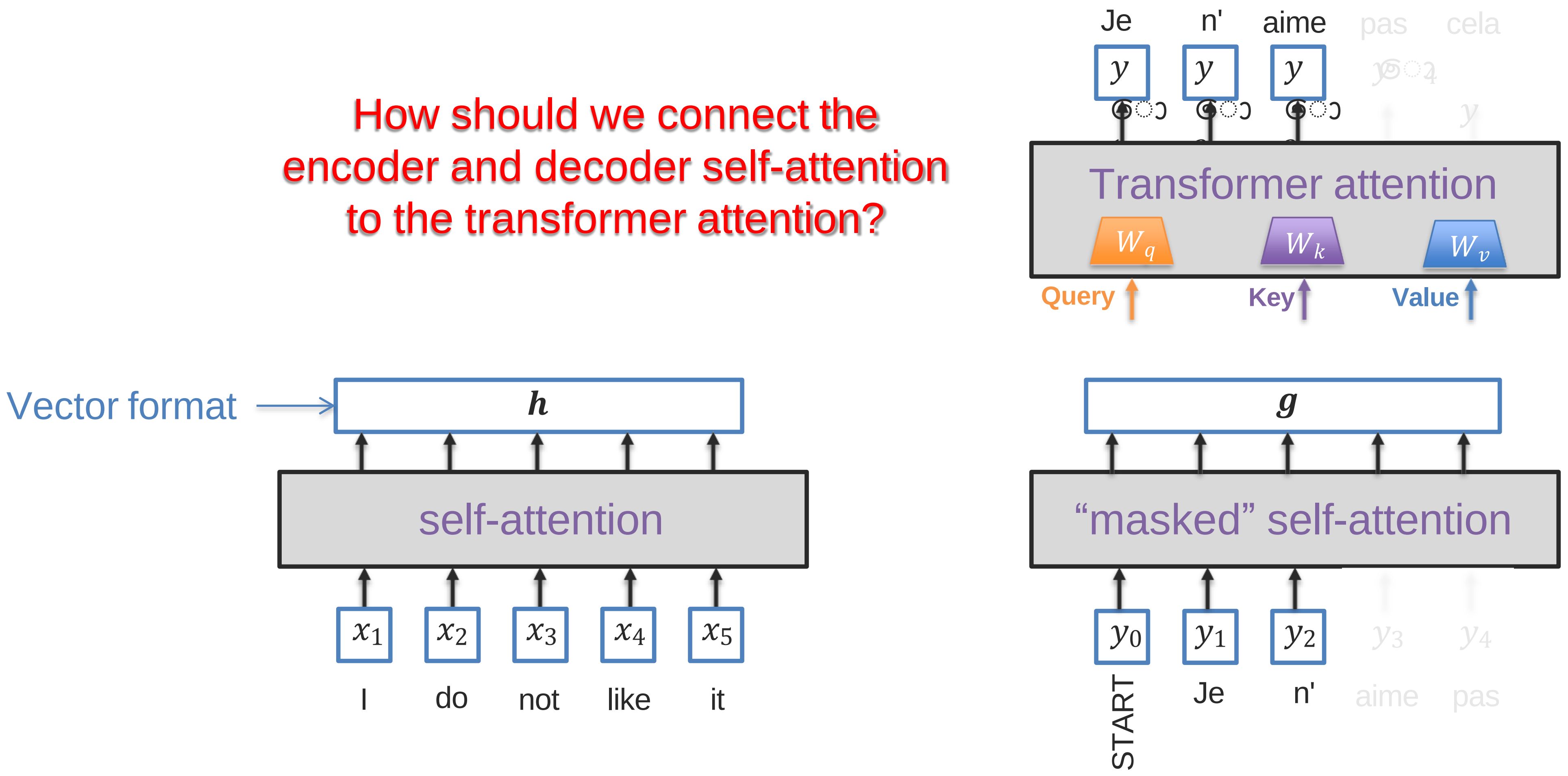


Seq2Seq with Transformer Attentions

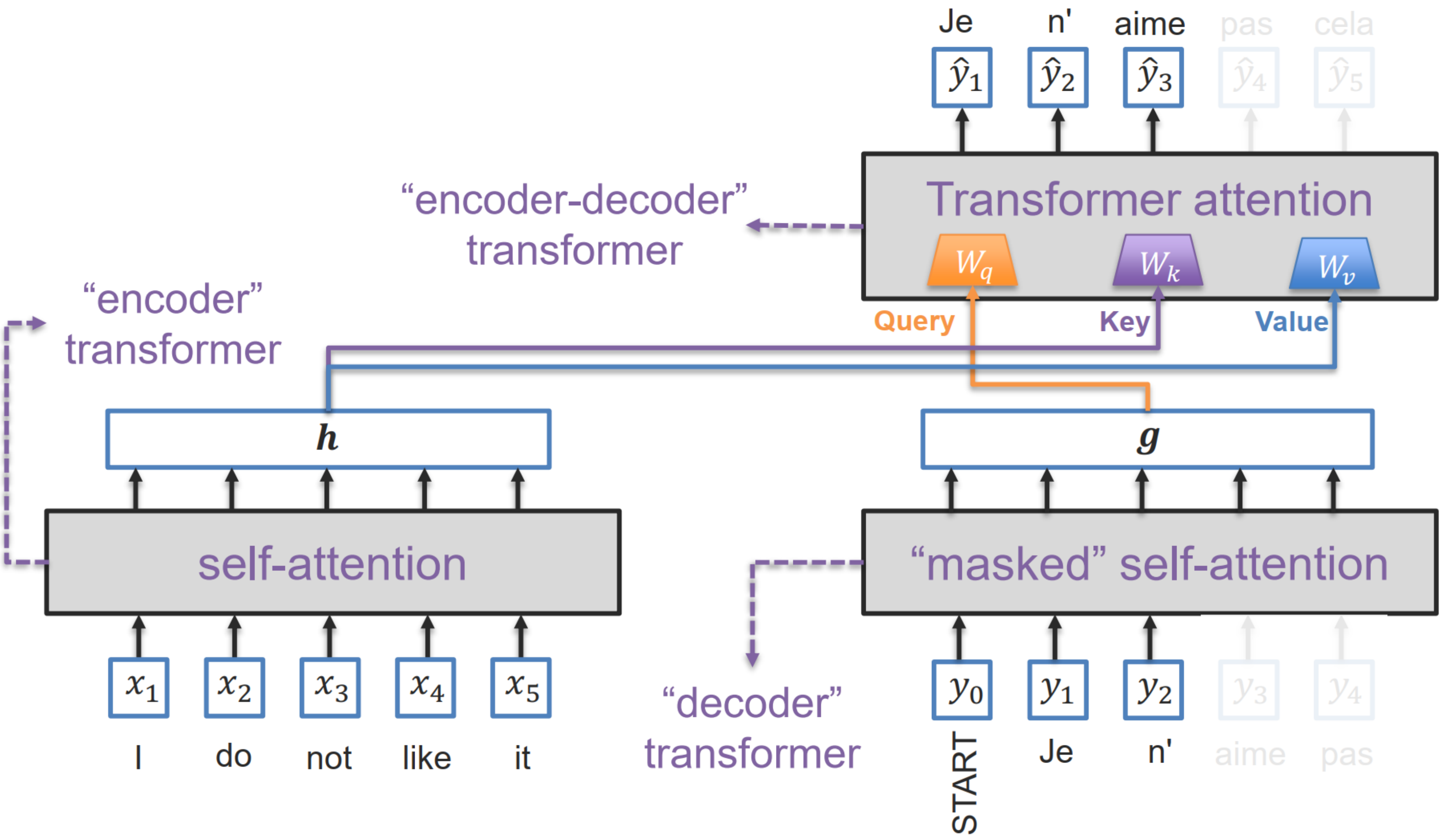


Seq2Seq with Transformer Attentions

How should we connect the encoder and decoder self-attention to the transformer attention?



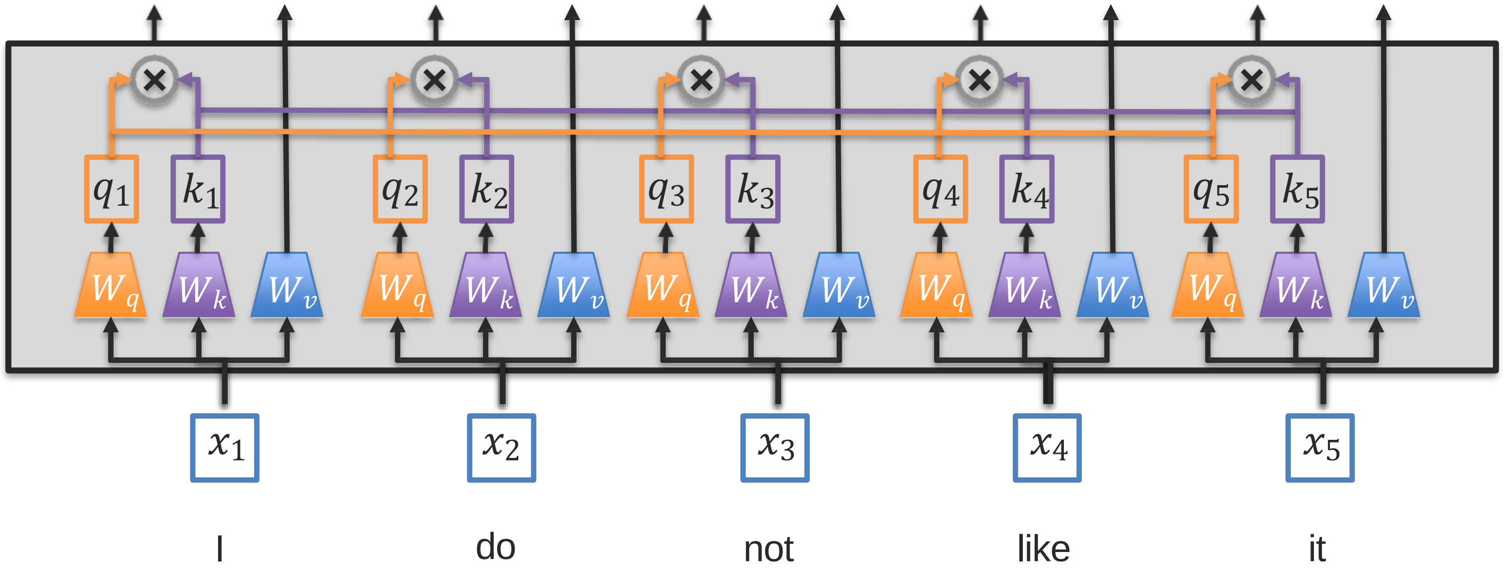
Seq2Seq with Transformer Attentions



Going Beyond Sequences: Graph Representations

*slides adapted from Leskovec, Representation Learning on Networks. WWW 2018

Transformers – Fully-Connected Sequences

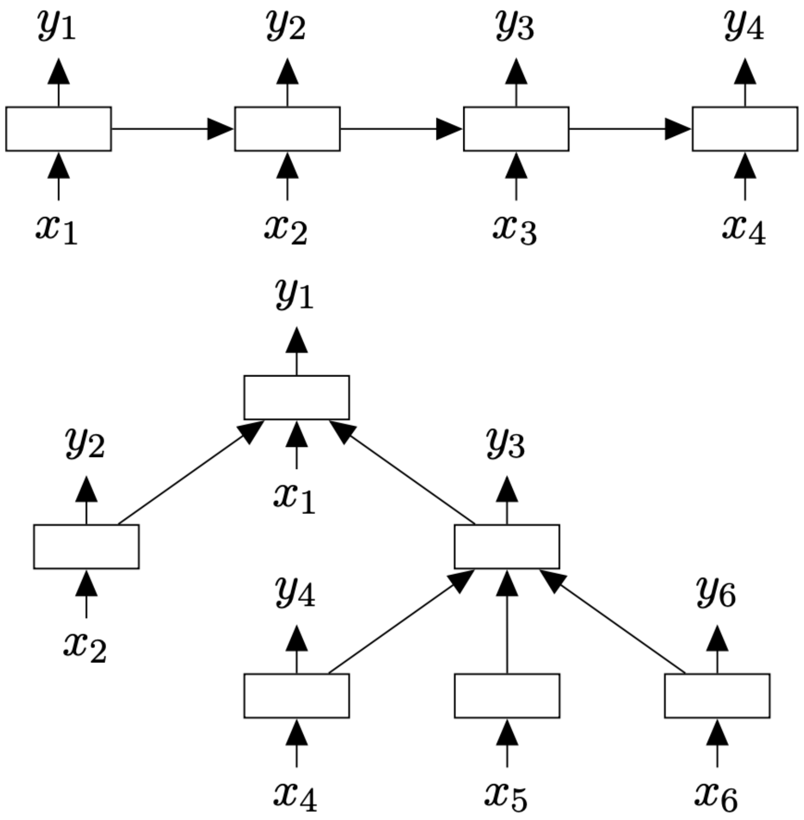


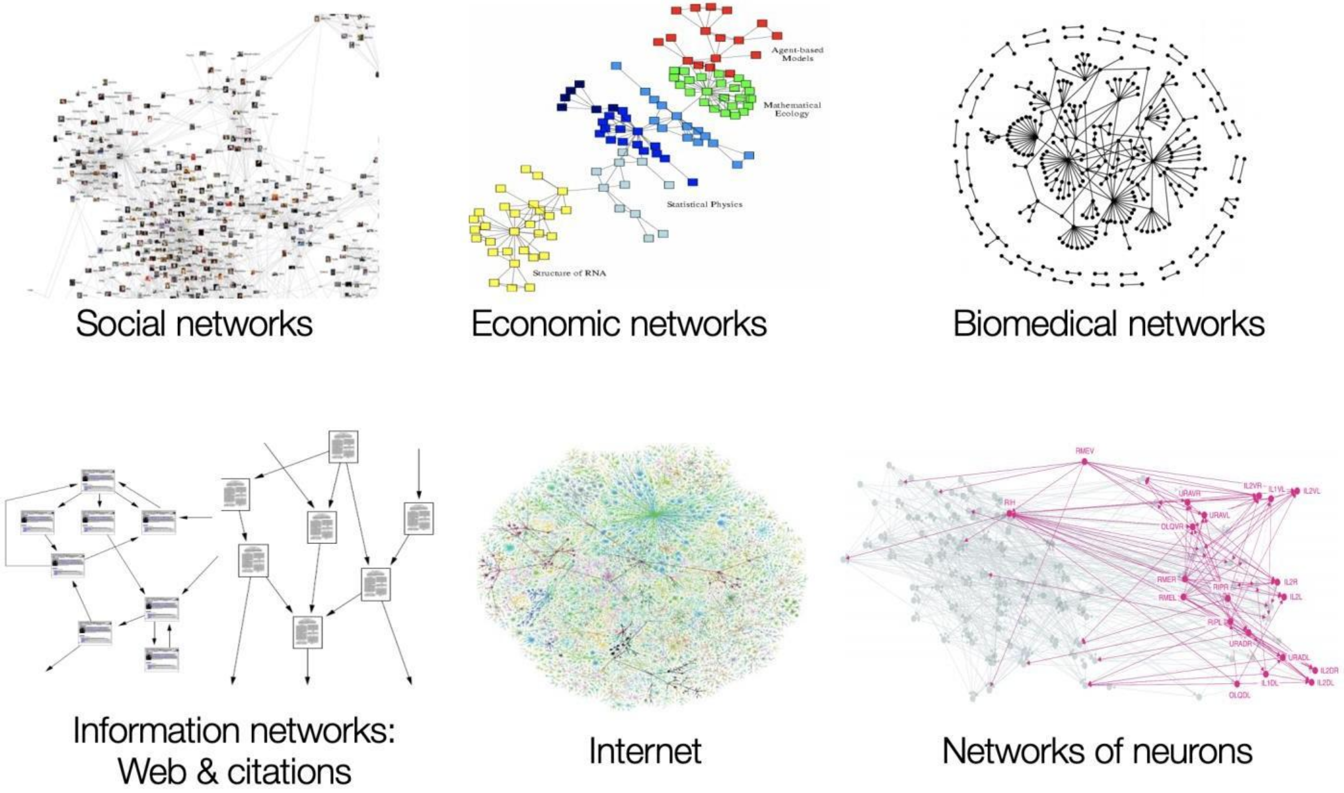
Should everything be connected to everything?

What if we have domain knowledge about connections?

Tree and Graph Networks

From linear chain models to tree and graph-structured models

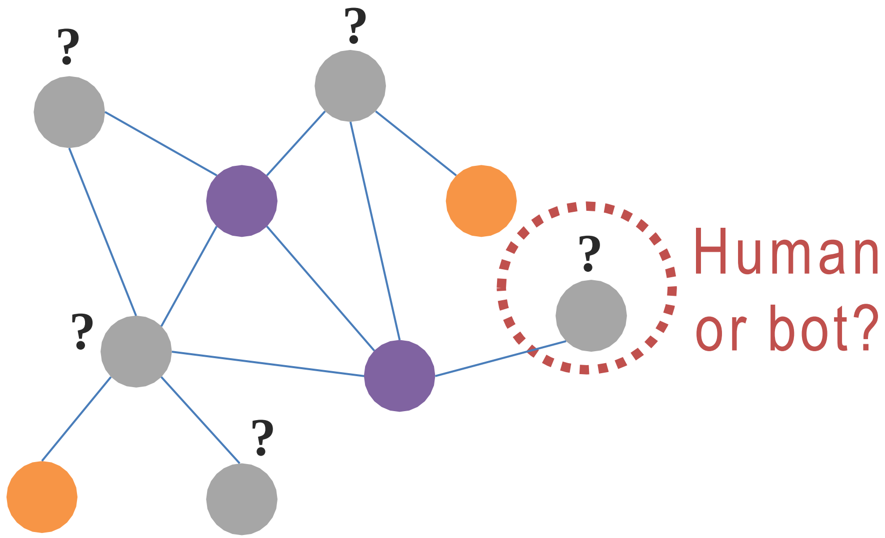




[Leskovec. Representation Learning on Networks. WWW 2018; Hamilton and Tang, Tutorial on Graph Representation Learning. AAAI 2019]

Graphs – Supervised Task

Goal: Learn from labels associated with a subset of nodes (or with all nodes)



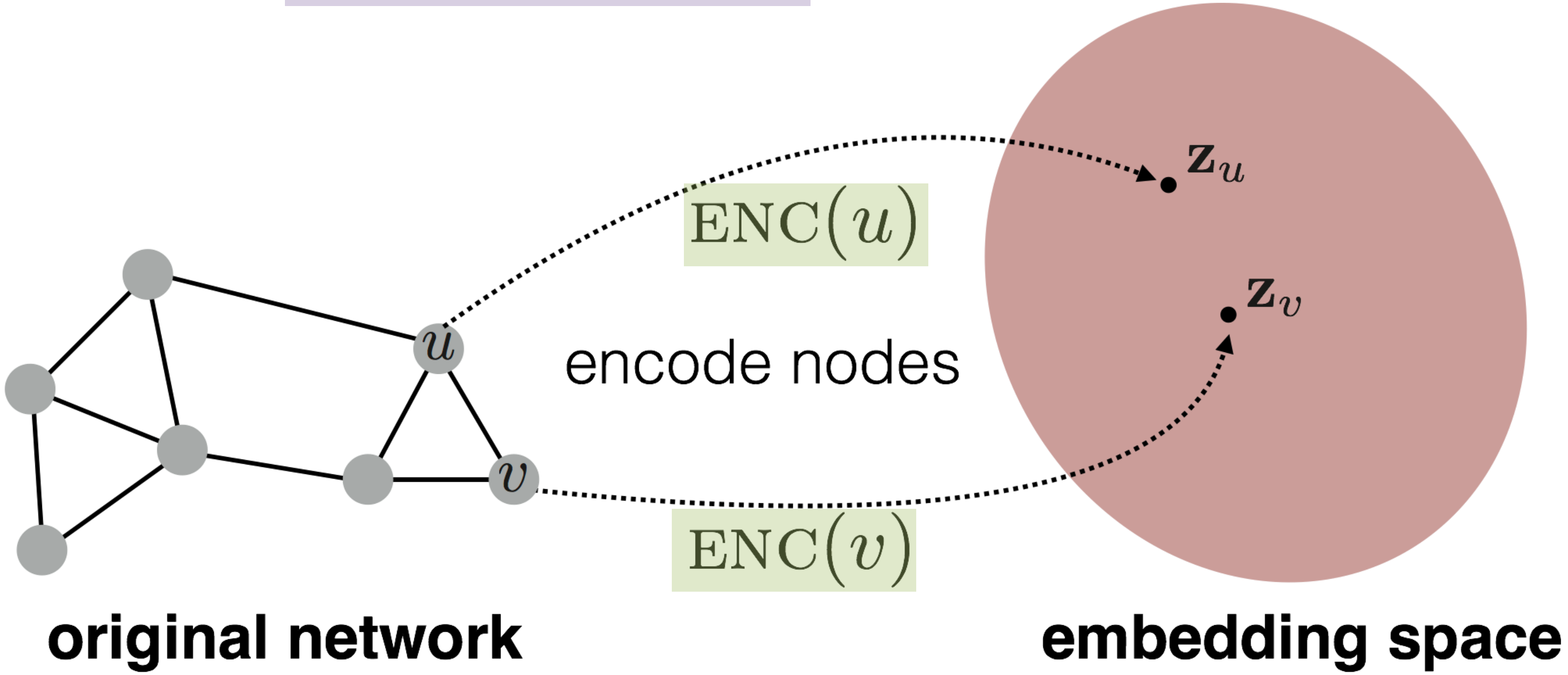
e.g., an online social network

[Leskovec. Representation Learning on Networks. WWW 2018; Hamilton and Tang, Tutorial on Graph Representation Learning. AAAI 2019]

Graphs – Unsupervised Task

Goal: Learn an embedding space where

$$\text{similarity}(u, v) \approx \mathbf{z}_v^\top \mathbf{z}_u$$



[Leskovec. Representation Learning on Networks. WWW 2018; Hamilton and Tang, Tutorial on Graph Representation Learning. AAAI 2019]

Graph Neural Nets

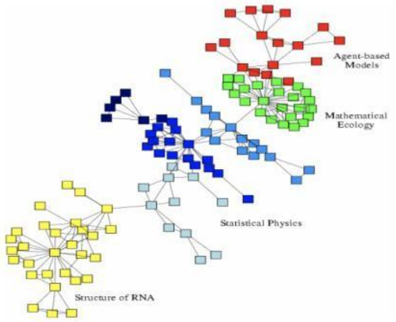
Assume we have a graph **G:**

V is the set of vertices
A is the binary adjacency matrix
X is a matrix of node features:

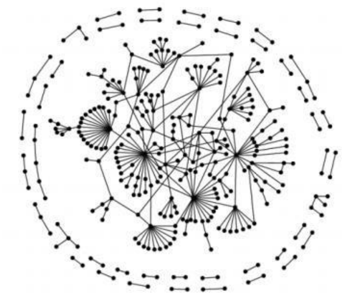
- Categorical attributes, text, image data
  e.g. profile information in a social network
- …

Y is a vector of node labels (optional)

Social networks



Economic networks

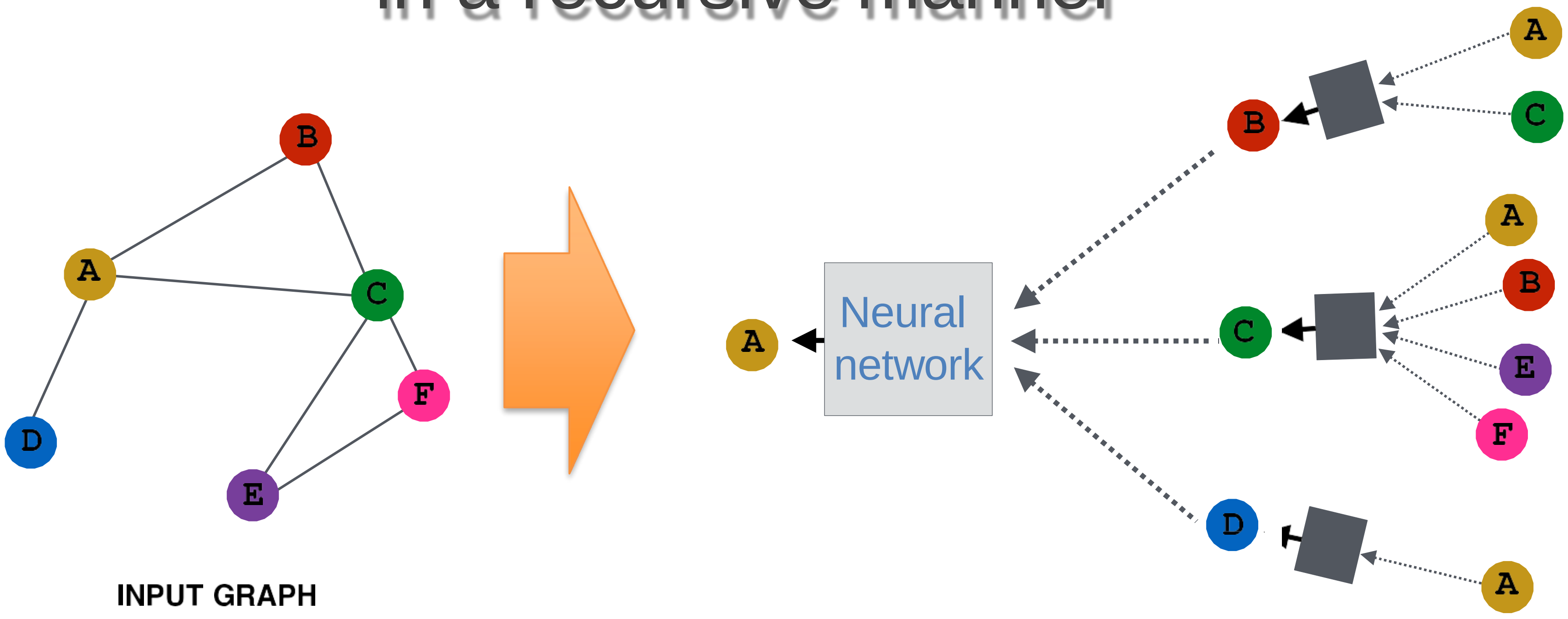


Biomedical networks

[Leskovec. Representation Learning on Networks. WWW 2018; Hamilton and Tang, Tutorial on Graph Representation Learning. AAAI 2019]

Graph Neural Nets

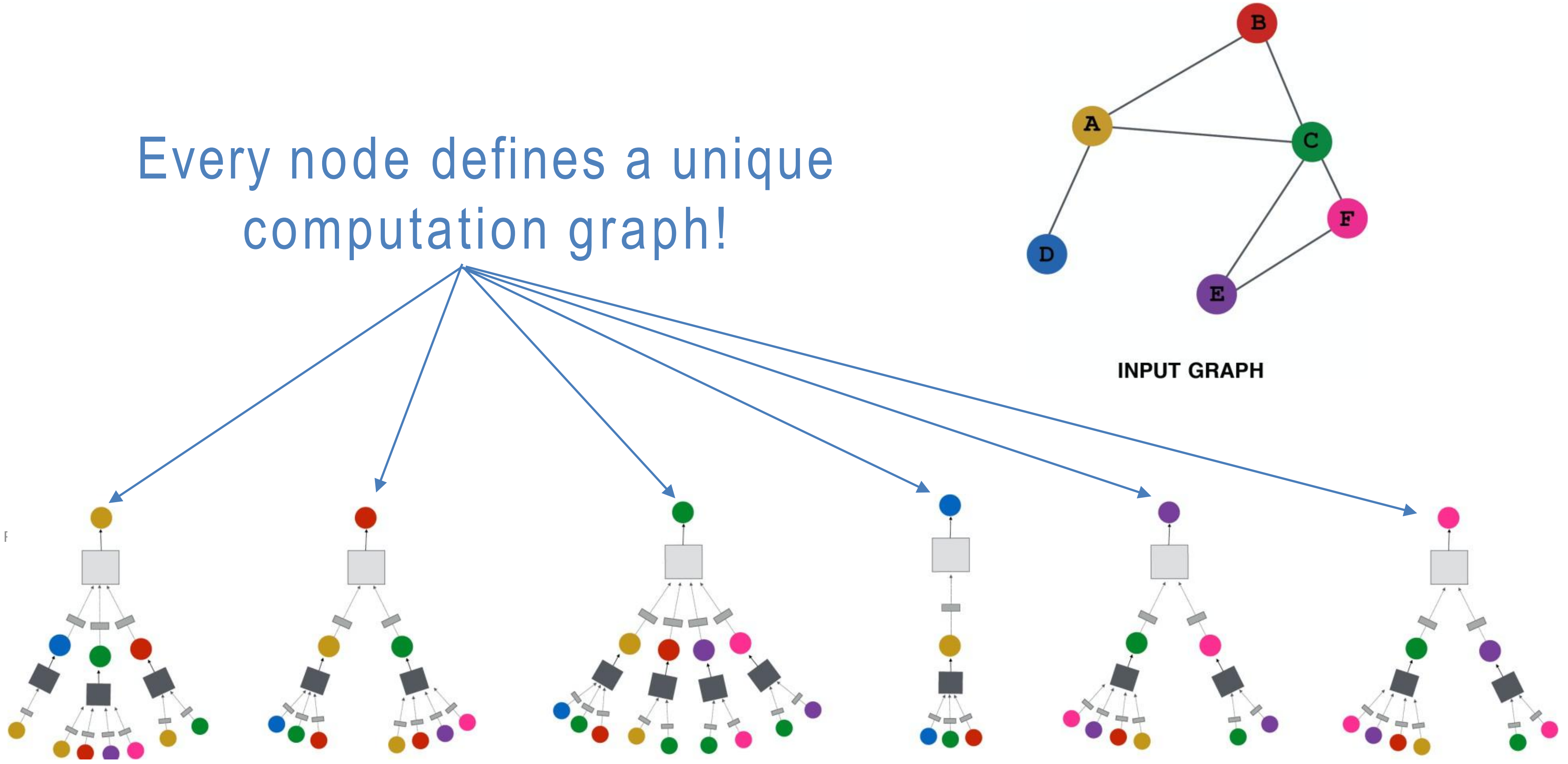
Key idea: Generate node embeddings based on local neighborhoods in a recursive manner



[Leskovec. Representation Learning on Networks. WWW 2018; Hamilton and Tang, Tutorial on Graph Representation Learning. AAAI 2019]

Graph Neural Nets

Every node defines a unique computation graph!

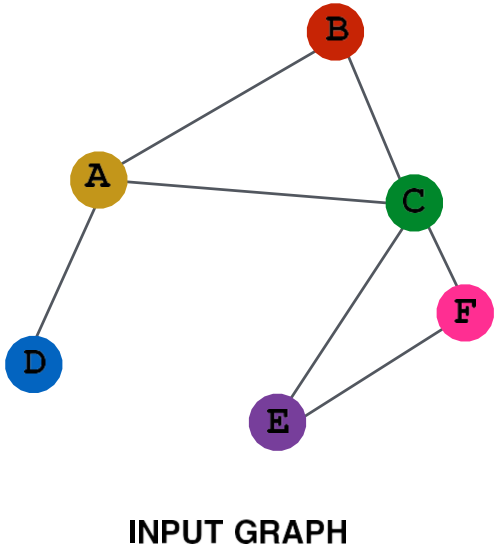


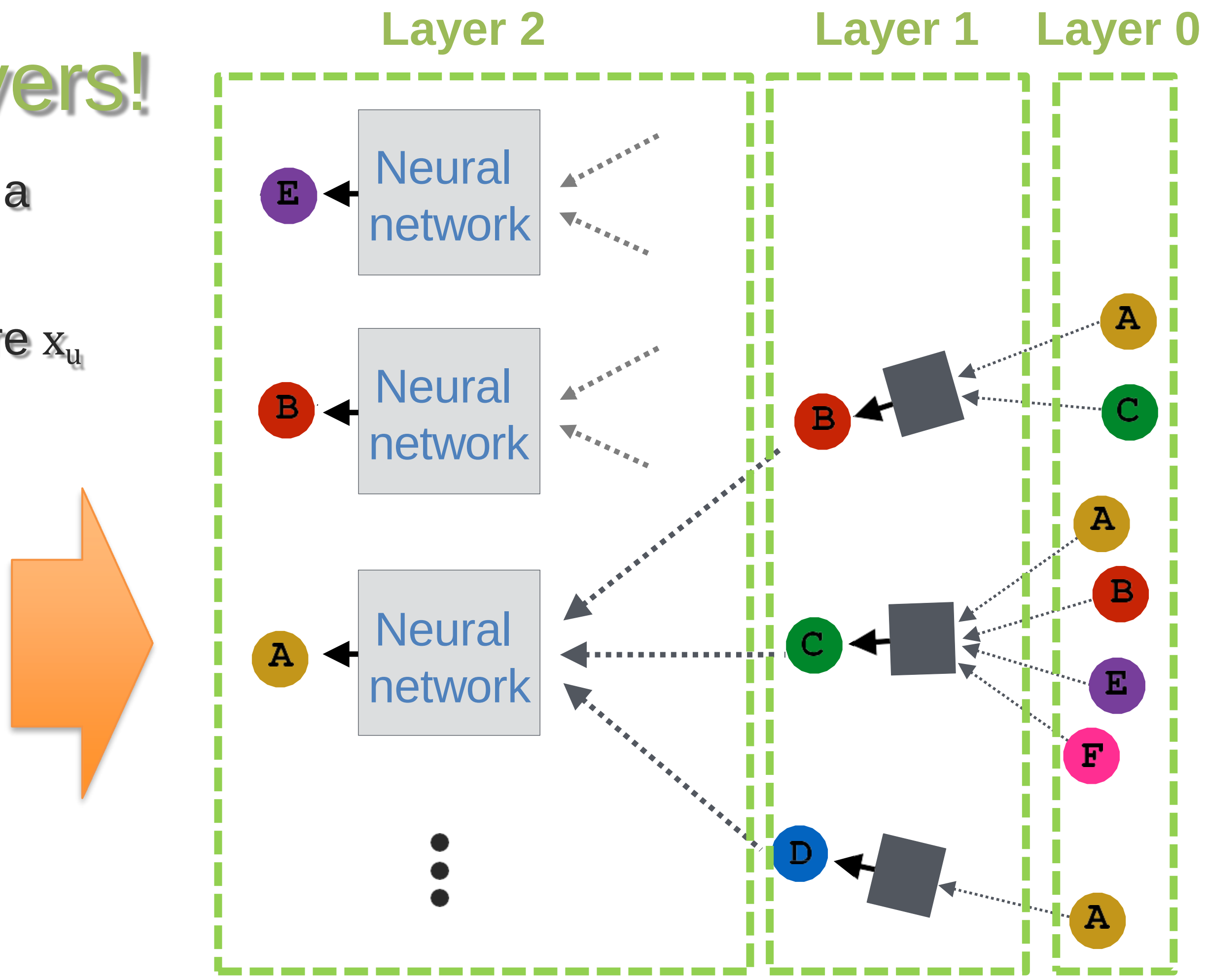
[Leskovec. Representation Learning on Networks. WWW 2018; Hamilton and Tang, Tutorial on Graph Representation Learning. AAAI 2019]

Graph Neural Nets

And multiple layers!

- Shared parameters within a specific layer
- "layer-0" is the input feature x_u





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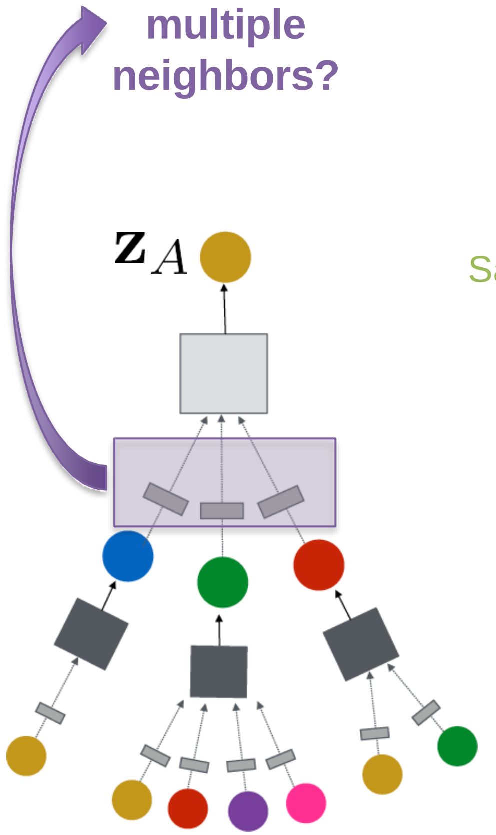
Graph Neural Nets – Neighborhood Aggregation



Average pooling (Scarselli et al., 2005)

$$\mathbf{h}_v^k = \sigma\left(\mathbf{W}_k \sum_{u \in N(v)} \frac{\mathbf{h}_u^{k-1}}{|N(v)|} + \mathbf{B}_k \mathbf{h}_v^{k-1}\right)$$

Different weights for neighbors and self

K is num layers

Graph Convolution Network (Kipf et al., 2017)

Same weights

$$\mathbf{h}_v^k = \sigma\left(\mathbf{W}_k \sum_{u \in N(v) \cup v} \frac{\mathbf{h}_u^{k-1}}{\sqrt{|N(u)||N(v)|}}\right)$$

Different normalization

- It can be efficiently implemented

Graph Attention Network (Velickovic et al., 2018)

Attention weights

$$\mathbf{h}_v^k = \sigma\left(\mathbf{W}_k \sum_{u \in N(v) \cup v} \frac{\alpha_{uv} \mathbf{h}_u^{k-1}}{\sqrt{|N(u)||N(v)|}}\right)$$

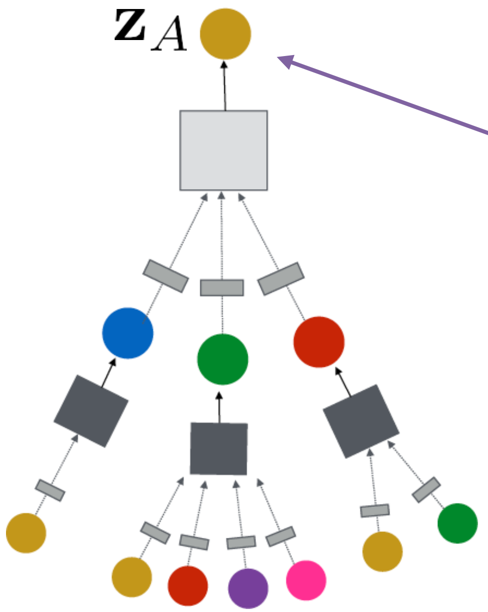
- Very similar to a self-attention transformer

[Leskovec. Representation Learning on Networks. WWW 2018; Hamilton and Tang, Tutorial on Graph Representation Learning. AAAI 2019]

Graph Neural Nets – Supervised Training

classification weights

$$\mathcal{L} = \sum_{v \in V} \boxed{y_v} \log(\sigma(\boxed{\mathbf{z}_v^\top} \boxed{\theta})) + (1 - \boxed{y_v}) \log(1 - \sigma(\boxed{\mathbf{z}_v^\top} \boxed{\theta}))$$



output node embedding

node class label

[Leskovec. Representation Learning on Networks. WWW 2018; Hamilton and Tang, Tutorial on Graph Representation Learning. AAAI 2019]

Going Beyond Sequences: Hierarchical Structure

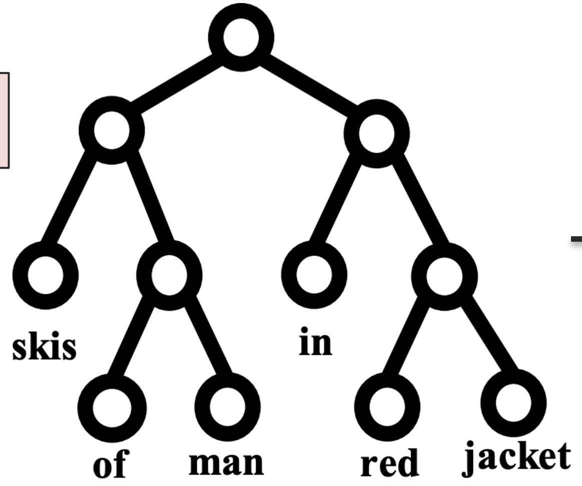
*slides adapted from Leskovec, Representation Learning on Networks. WWW 2018

Hierarchical Structure

Leverage syntactic structure of language

Parse

Skis of man in red jacket.



Object detection

[Hong et al., Learning to Compose and Reason with Language Tree Structures for Visual Grounding. IEEE TPAMI 2019]

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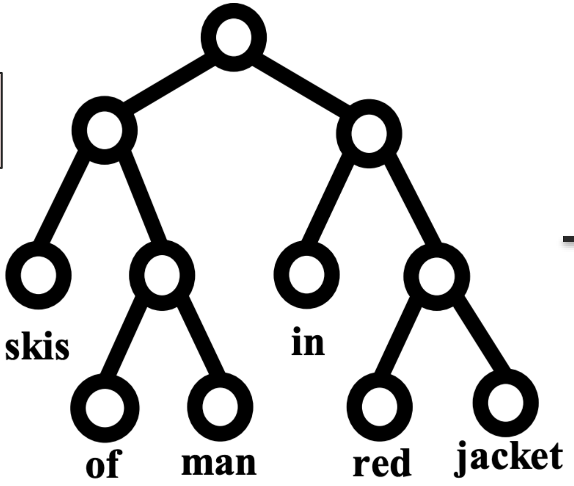
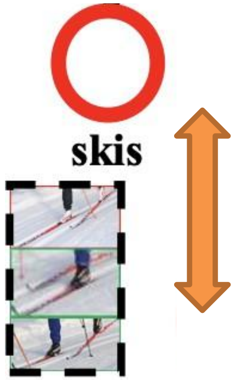


Object detection

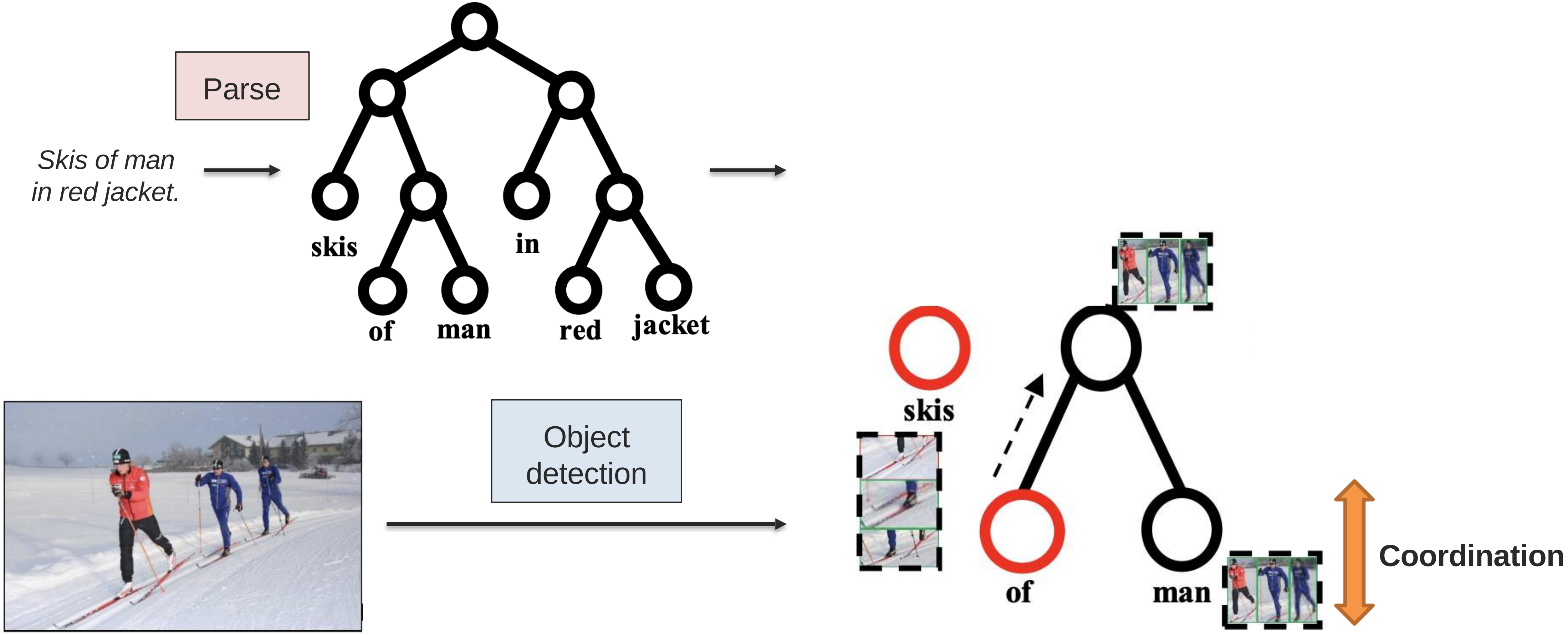


Coordination

[Hong et al., Learning to Compose and Reason with Language Tree Structures for Visual Grounding. IEEE TPAMI 2019]

Hierarchical Structure

Leverage syntactic structure of language



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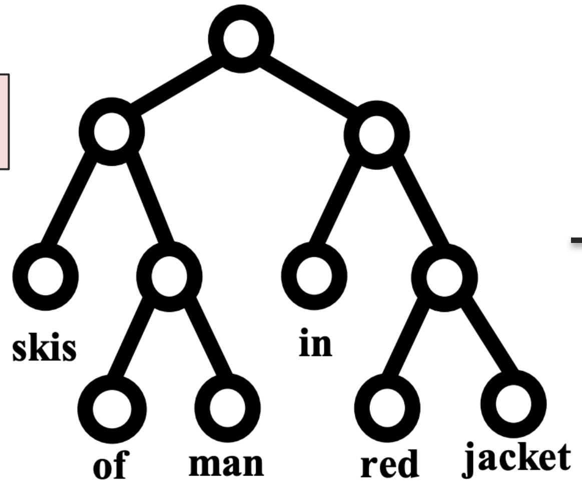

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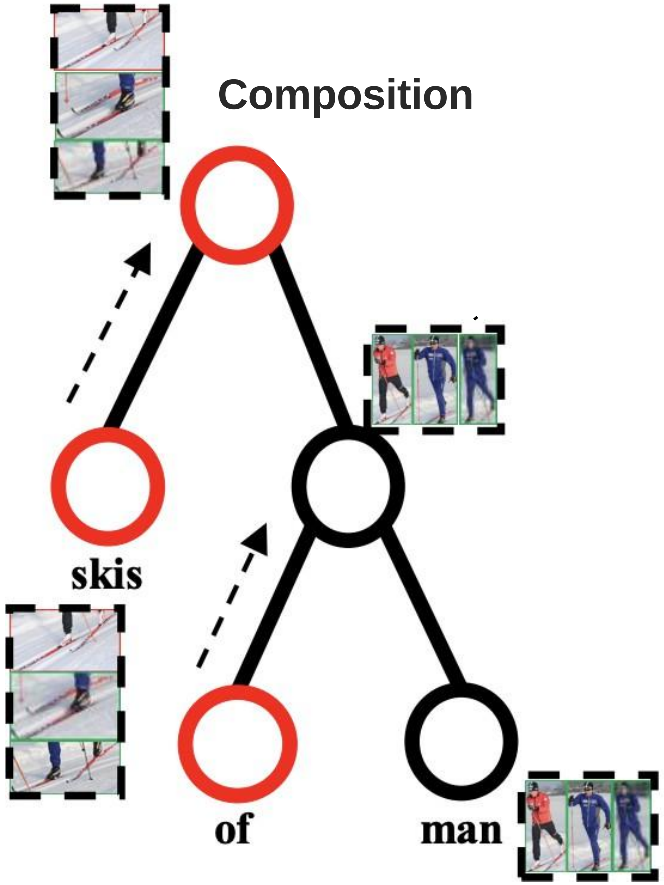

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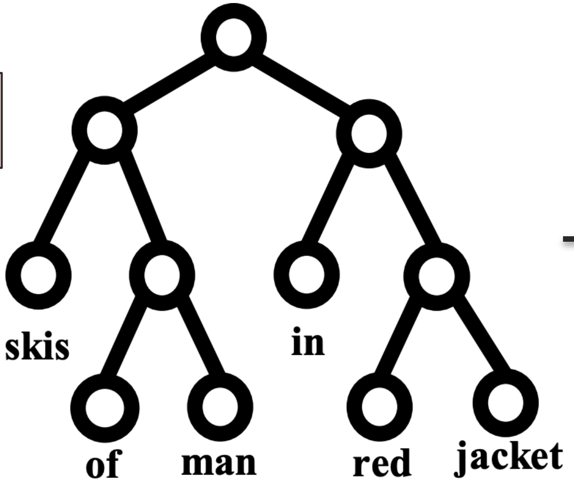

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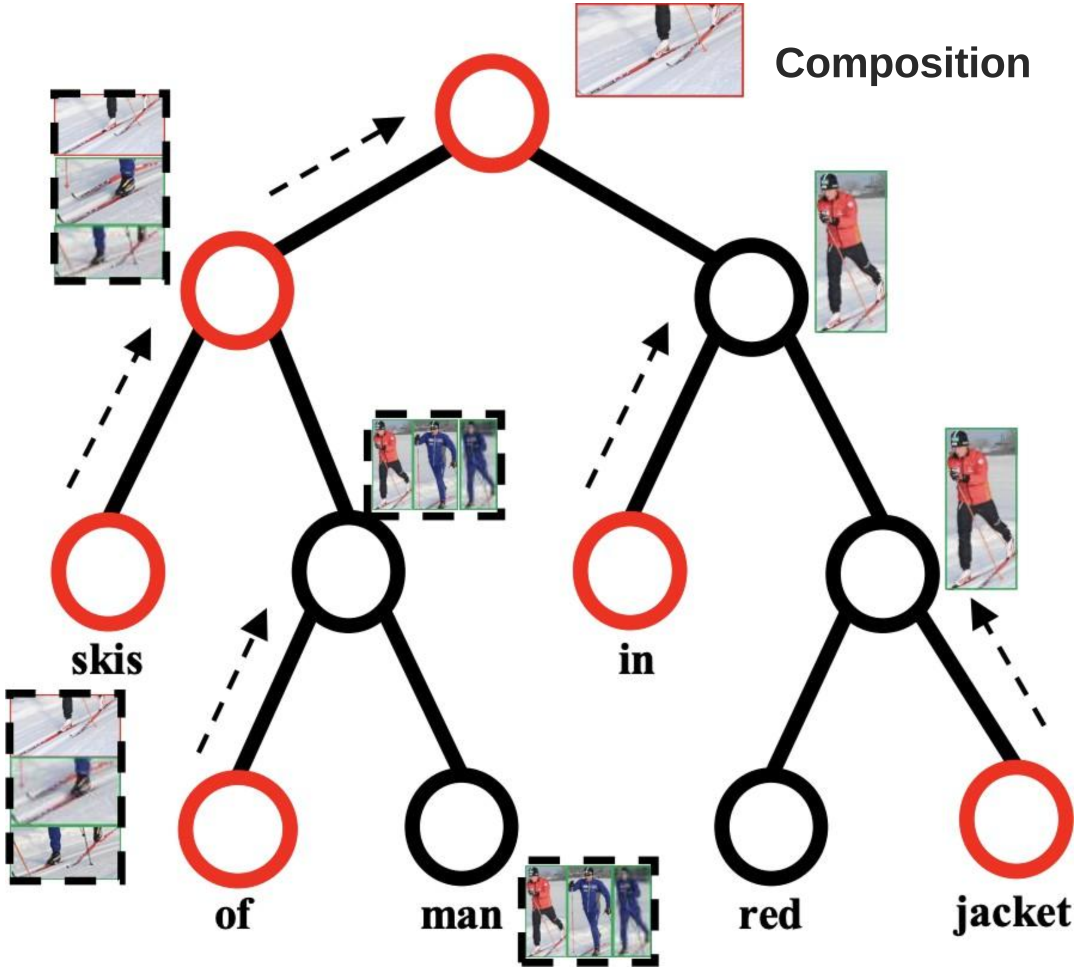

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Hierarchical Structure

Leverage syntactic structure of language

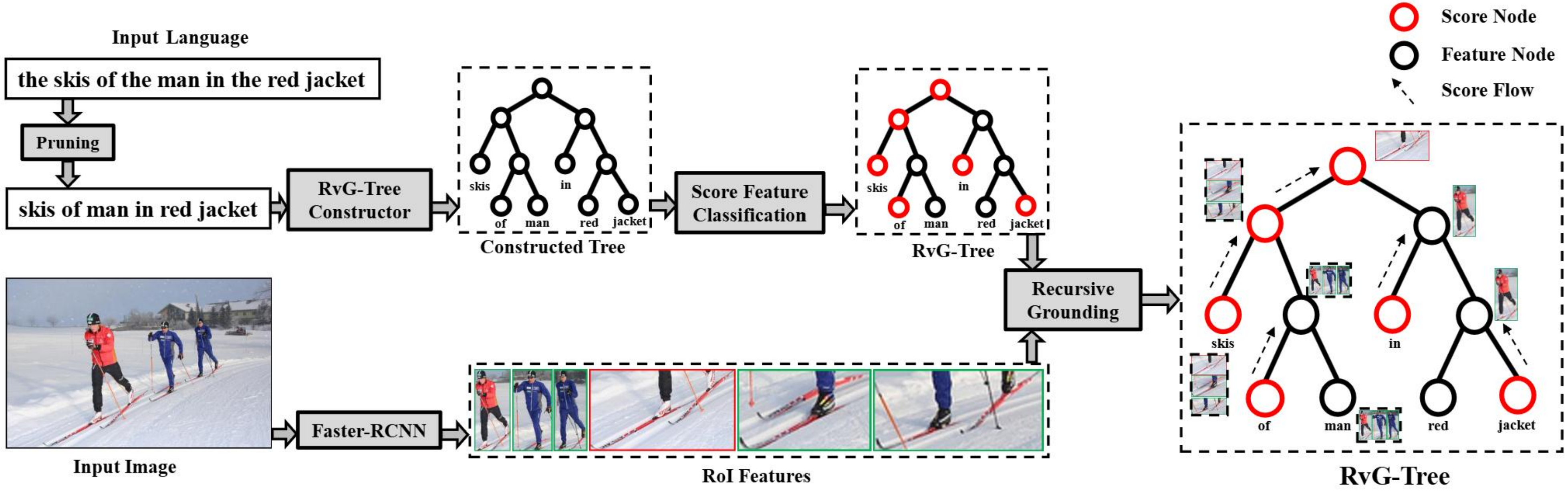


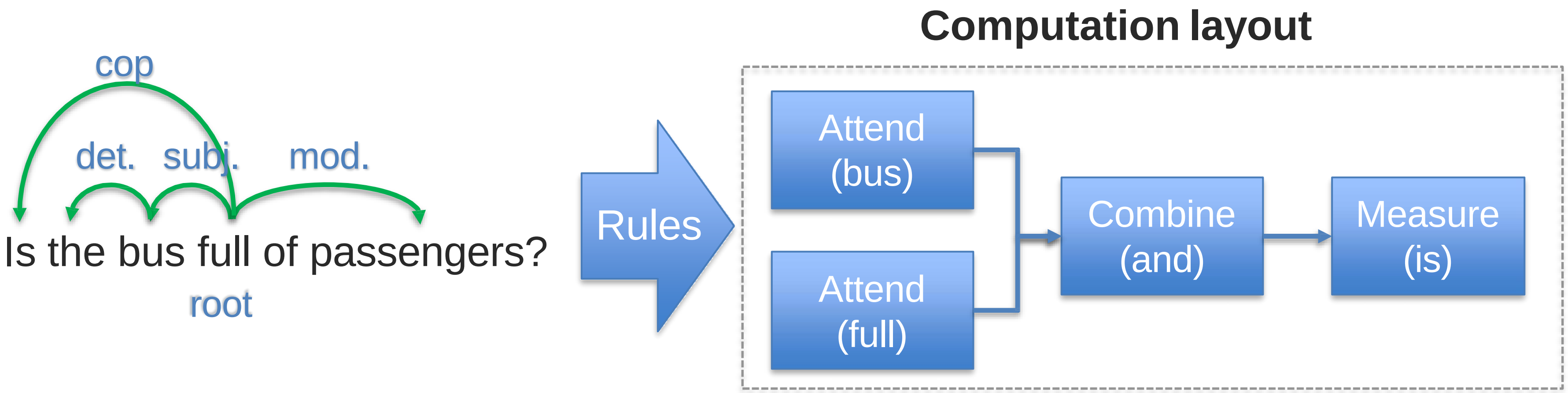
Fig. 2: The overview of using RvG-Tree for natural language grounding. Given a natural language sentence as input, we first prune the sentence and then construct RvG-Tree (Section 3.2). Then, we have a score-feature classifier (Section 3.3) to determine each node as the "score node" or "feature node", where the score node returns the recursive score and the feature node returns the feature (Section 3.3). The final score of the root node is accumulated recursively in a bottom-up fashion (Section 3.3) and the visual region with the highest score is considered as the grounding result. Note that all the nodes can be visualized by the corresponding confidence scores and only qualitative regions are visualized.

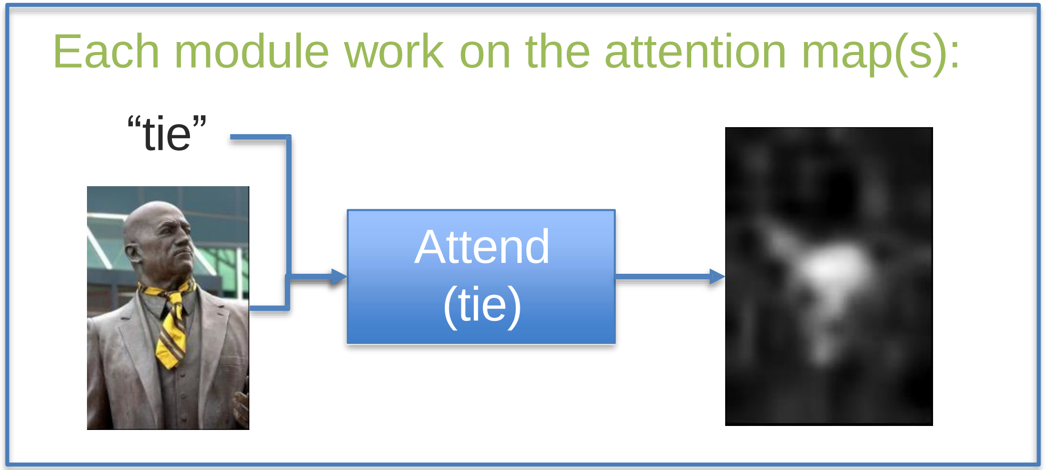
[Hong et al., Learning to Compose and Reason with Language Tree Structures for Visual Grounding. IEEE TPAMI 2019]

Going Beyond Sequences: Modular Structure

*slides adapted from Leskovec, Representation Learning on Networks. WWW 2018

Neural Module Network



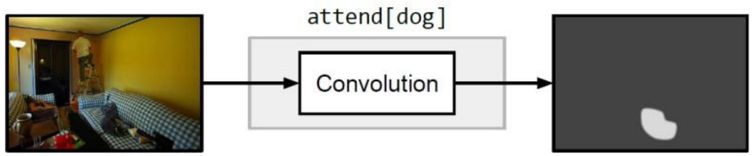


Andreas et al., Deep Compositional Question Answering with Neural Module Networks, 2016

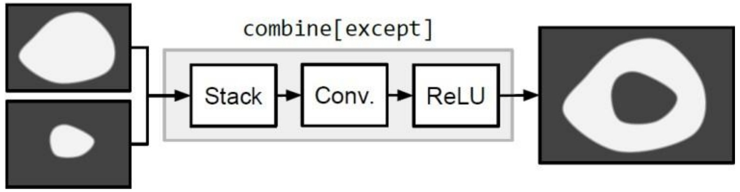
Predefined Set of Modules

1) Analyze the image:

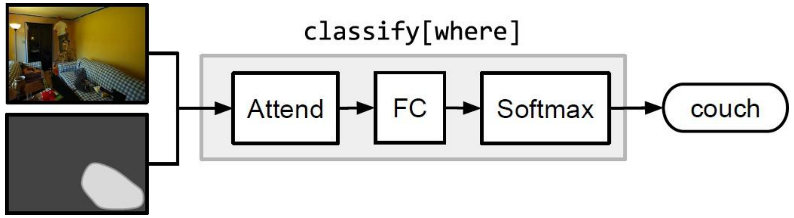
attend : $Image \rightarrow Attention$



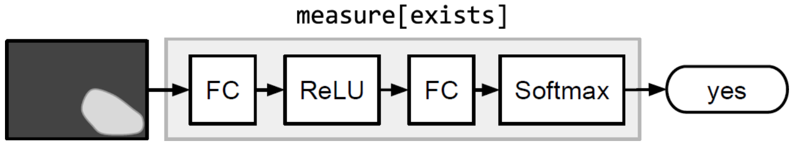
combine : $Attention \times Attention \rightarrow Attention$



2) Make a prediction



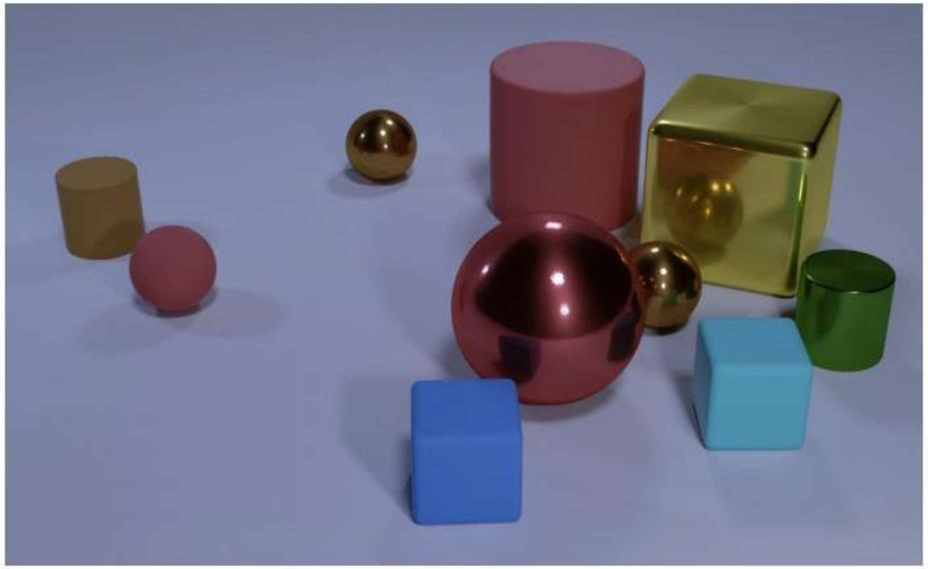
measure : $Attention \rightarrow Label$



Andreas et al., Deep Compositional Question Answering with Neural Module Networks, 2016

CLEVR: Dataset for Visual Reasoning

Perfect for a neural module network!



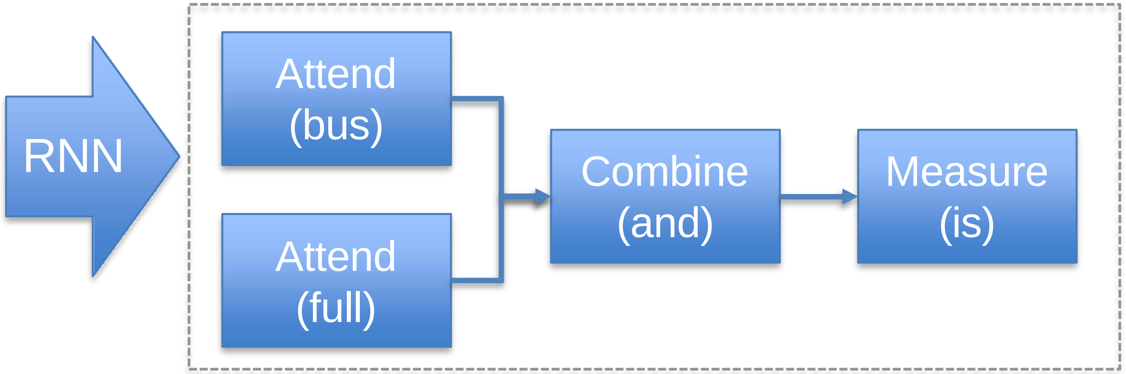
Q: Are there an equal number of large things and metal spheres? **Q:** What size is the cylinder that is left of the brown metal thing that is left of the big sphere? **Q:** There is a sphere with the same size as the metal cube; is it made of the same material as the small red sphere? **Q:** How many objects are either small cylinders or metal things?

Johnson et al., CLEVR: A Diagnostic Dataset for Compositional Language and Elementary Visual Reasoning, CVPR 2017

Module Network V2: End-to-End Learning

Computation layout

Is the bus full of passengers?



No need to parse the question!

No rule-based creation of the layout!

Hu et al., Learning to Reason: End-to-End Module Networks for Visual Question Answering, 2017

Module Network V2: End-to-End Learning

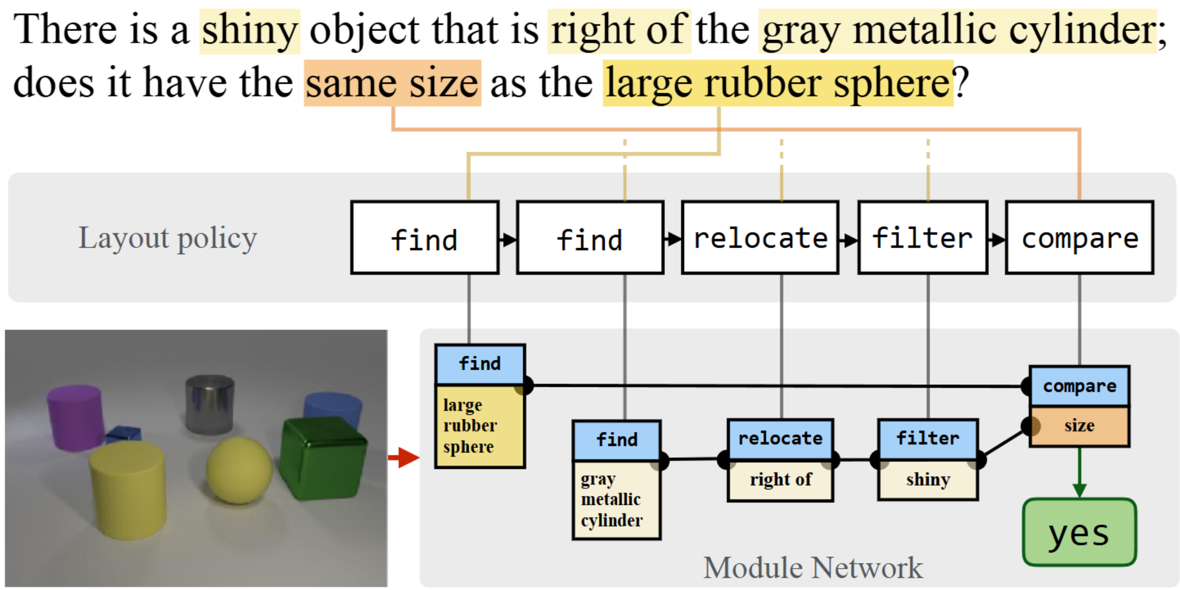
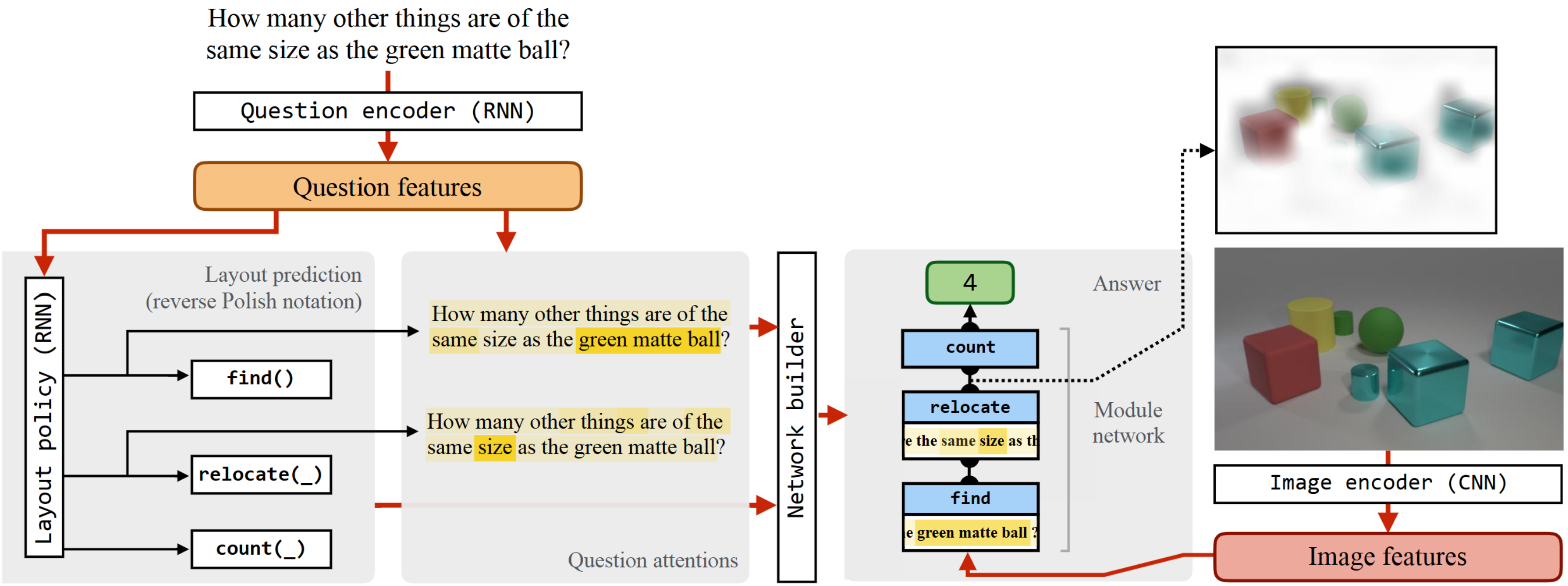


Figure 1: For each instance, our model predicts a computational expression and a sequence of attentive module parameterizations. It uses these to assemble a concrete network architecture, and then executes the assembled neural module network to output an answer for visual question answering. (The example shows a real structure predicted by our model, with text attention maps simplified for clarity.)

Hu et al., Learning to Reason: End-to-End Module Networks for Visual Question Answering, 2017

Module Network V2: End-to-End Learning



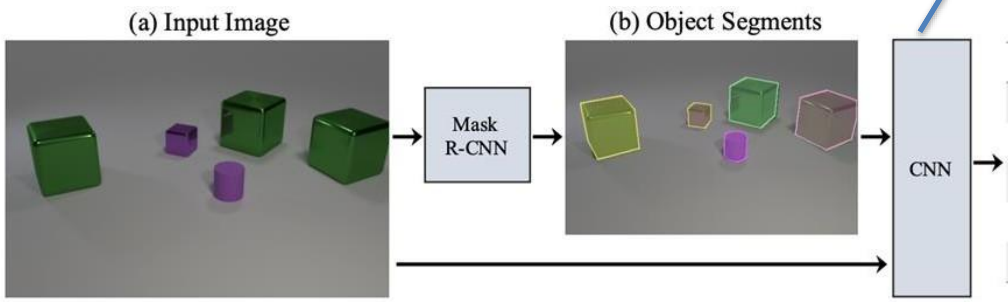
Model overview. Our approach first computes a deep representation of the question, and uses this as an input to a layout-prediction policy implemented with a recurrent neural network. This policy emits both a sequence of structural actions, specifying a template for a modular neural network in reverse Polish notation, and a sequence of attentive actions, extracting parameters for these neural modules from the input sentence. These two sequences are passed to a network builder, which dynamically instantiates an appropriate neural network and applies it to the input image to obtain an answer.

Hu et al., Learning to Reason: End-to-End Module Networks for Visual Question Answering, 2017

Module Network V3: Neural-symbolic VQA

1) Image Attributes

Previously trained in a supervised way



(c) Structural Scene Representation

ID	Size	Shape	Material	Color	x	y	z
1	Small	Cube	Metal	Purple	-0.45	-1.10	0.35
2	Large	Cube	Metal	Green	3.83	-0.04	0.70
3	Large	Cube	Metal	Green	-3.20	0.63	0.70
4	Small	Cylinder	Rubber	Purple	0.75	1.31	0.35
5	Large	Cube	Metal	Green	1.58	-1.60	0.70

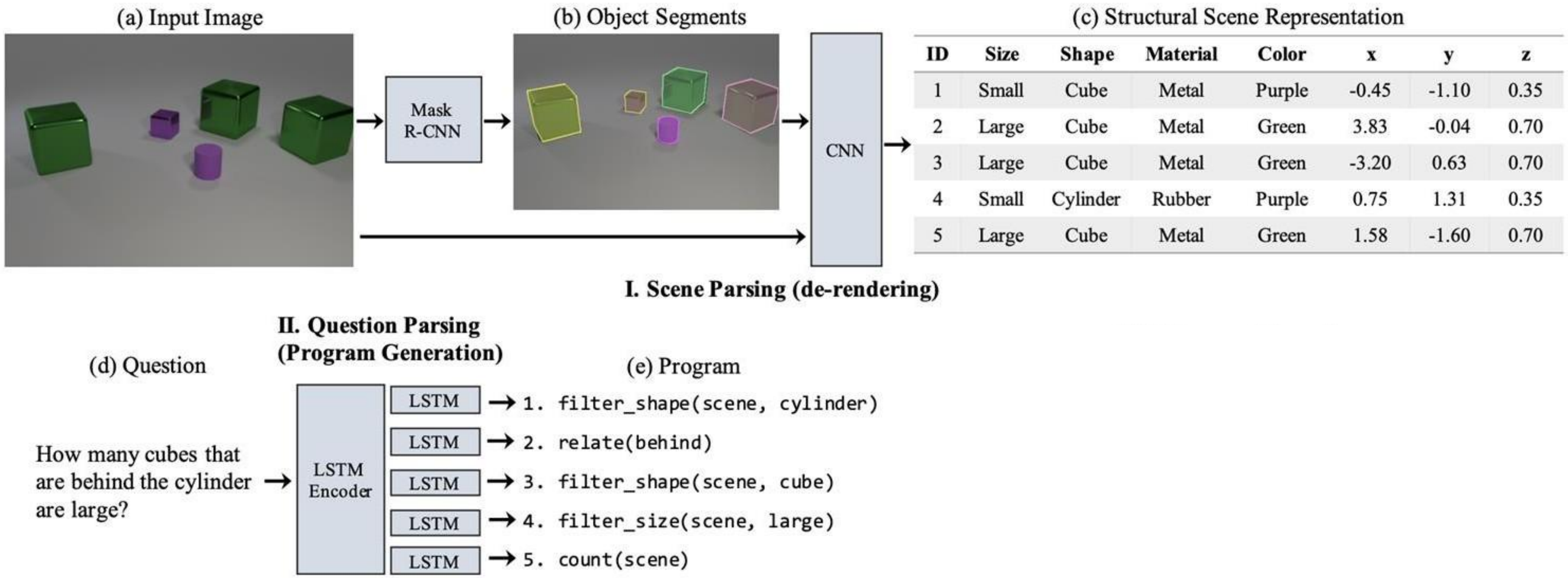
I. Scene Parsing (de-rendering)

Kexin Yi, et al. “Neural-Symbolic VQA: Disentangling Reasoning from Vision and Language Understanding.” Neurips 2018

Module Network V3: Neural-symbolic VQA

2) Parsing questions into programs

Similar to neural module networsk



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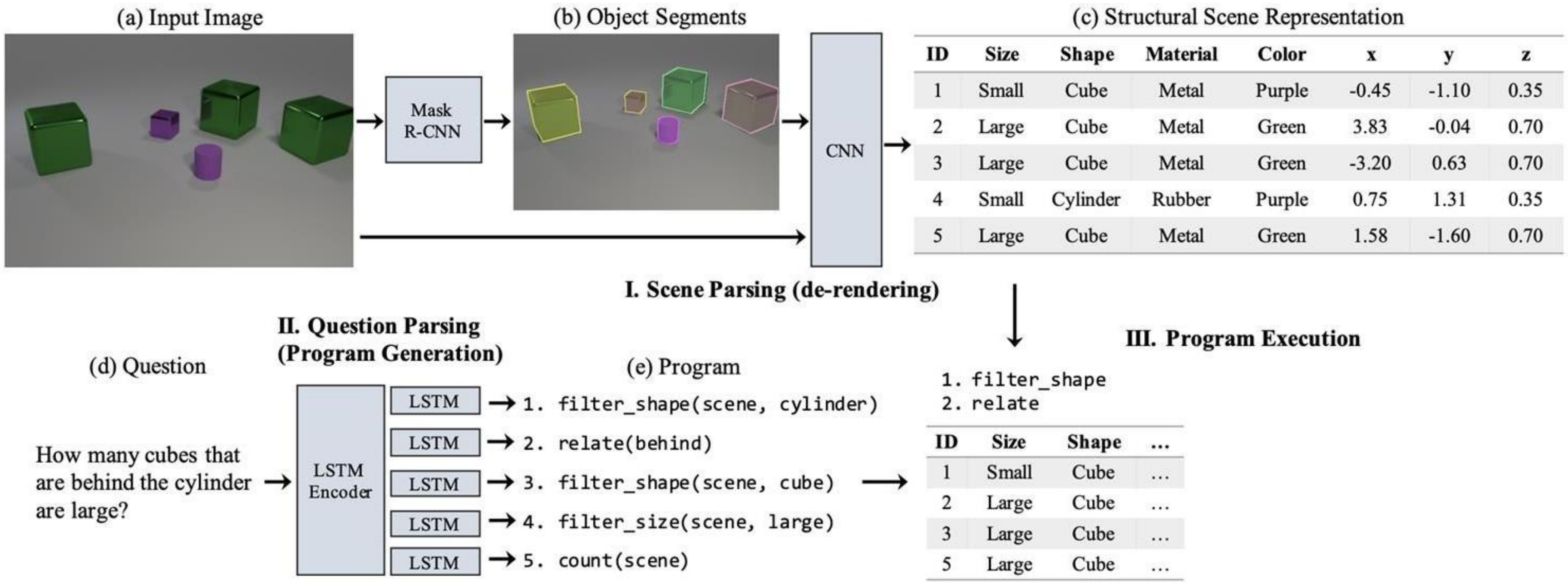
Kexin Yi, et al. “Neural-Symbolic VQA: Disentangling Reasoning from Vision and Language Understanding.” Neurips 2018

Module Network V3: Neural-symbolic VQA

3) Program execution

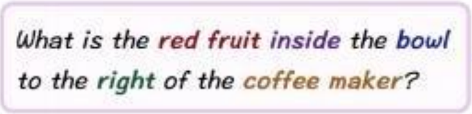
Execution of the program is somewhat easier given the “symbolic” representation of the image



Kexin Yi, et al. “Neural-Symbolic VQA: Disentangling Reasoning from Vision and Language Understanding.” Neurips 2018

Module Networks V4: The Neural State Machine

How to solve this question using visual reasoning?



1. Given an **image**, generate a probabilistic **scene graph** that captures the semantic concepts.

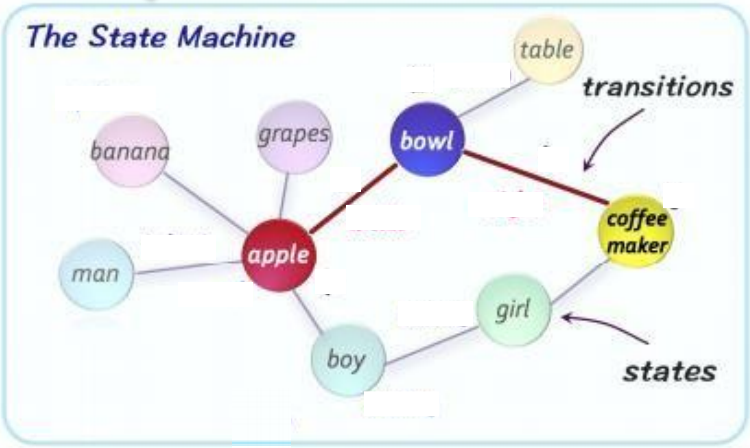
2. Treat the graph as a **state machine** and simulate iterative computation over it to *answer questions* or *draw inferences*.

3. Natural language questions are translated into *soft instructions* and used to perform sequential reasoning over the scene graph/state machine.

Hudson, Drew, and Christopher D. Manning. "Learning by abstraction: The neural state machine.“ NeurIPS 2019

Module Networks V4: The Neural State Machine

Detect objects and create proximity graph

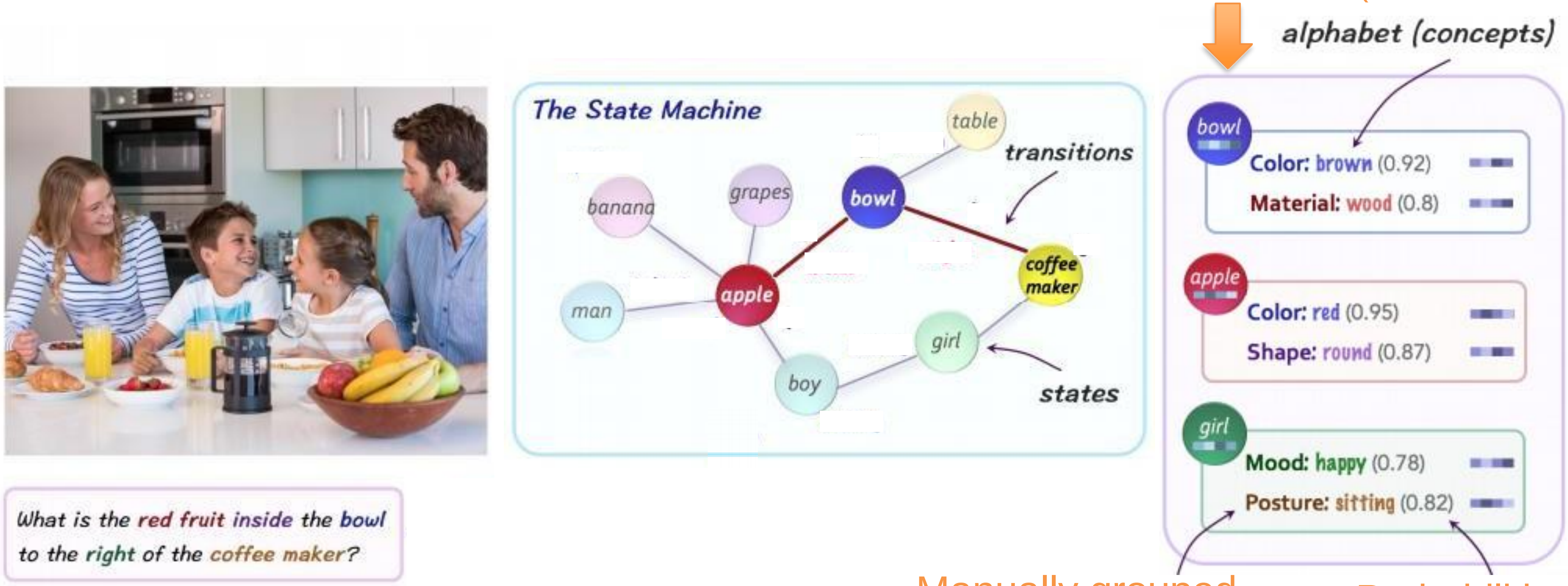


What is the *red fruit inside* the *bowl* to the *right* of the *coffee maker*?

Hudson, Drew, and Christopher D. Manning. "Learning by abstraction: The neural state machine.“ NeurIPS 2019

Module Networks V4: The Neural State Machine

Pre-trained an alphabet of concepts
(Visual Genome)



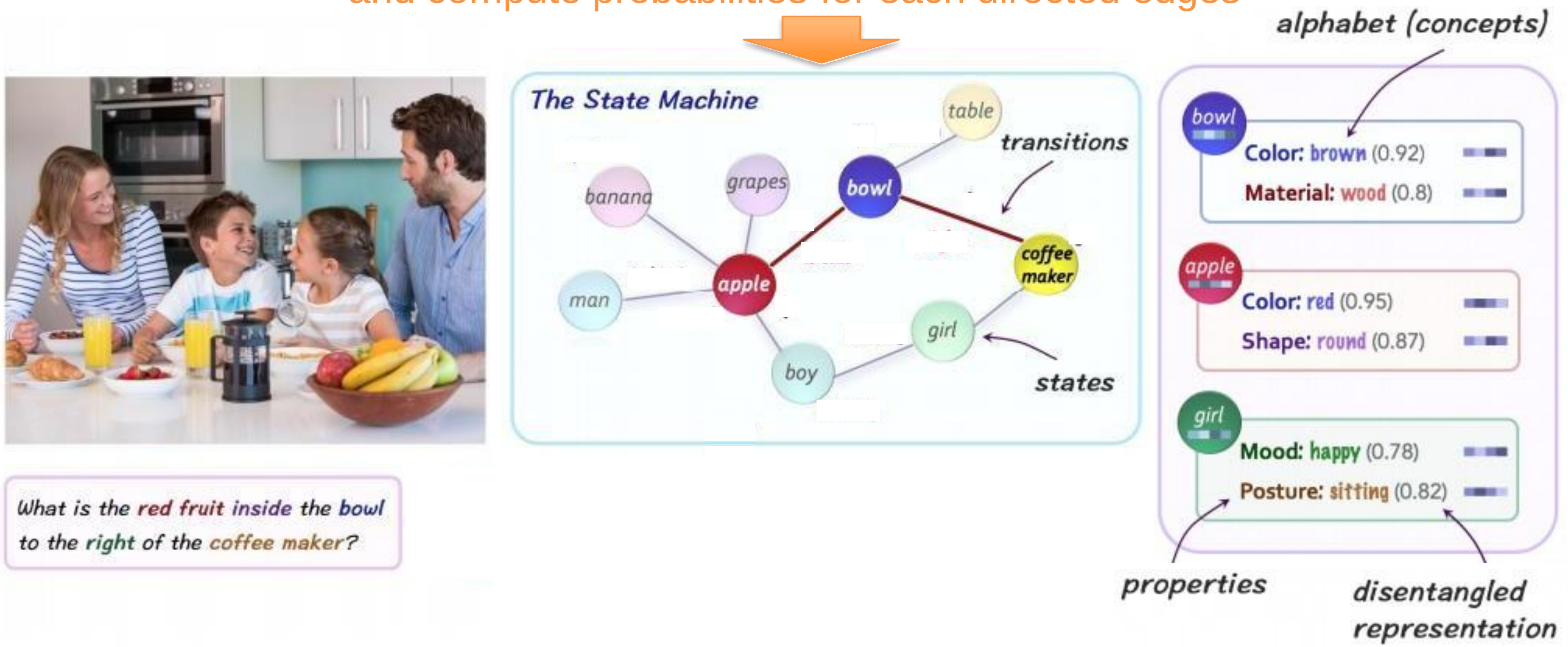
Manually grouped by "properties"

Probabilities computed at runtime for each object instance

Hudson, Drew, and Christopher D. Manning. "Learning by abstraction: The neural state machine." NeurIPS 2019

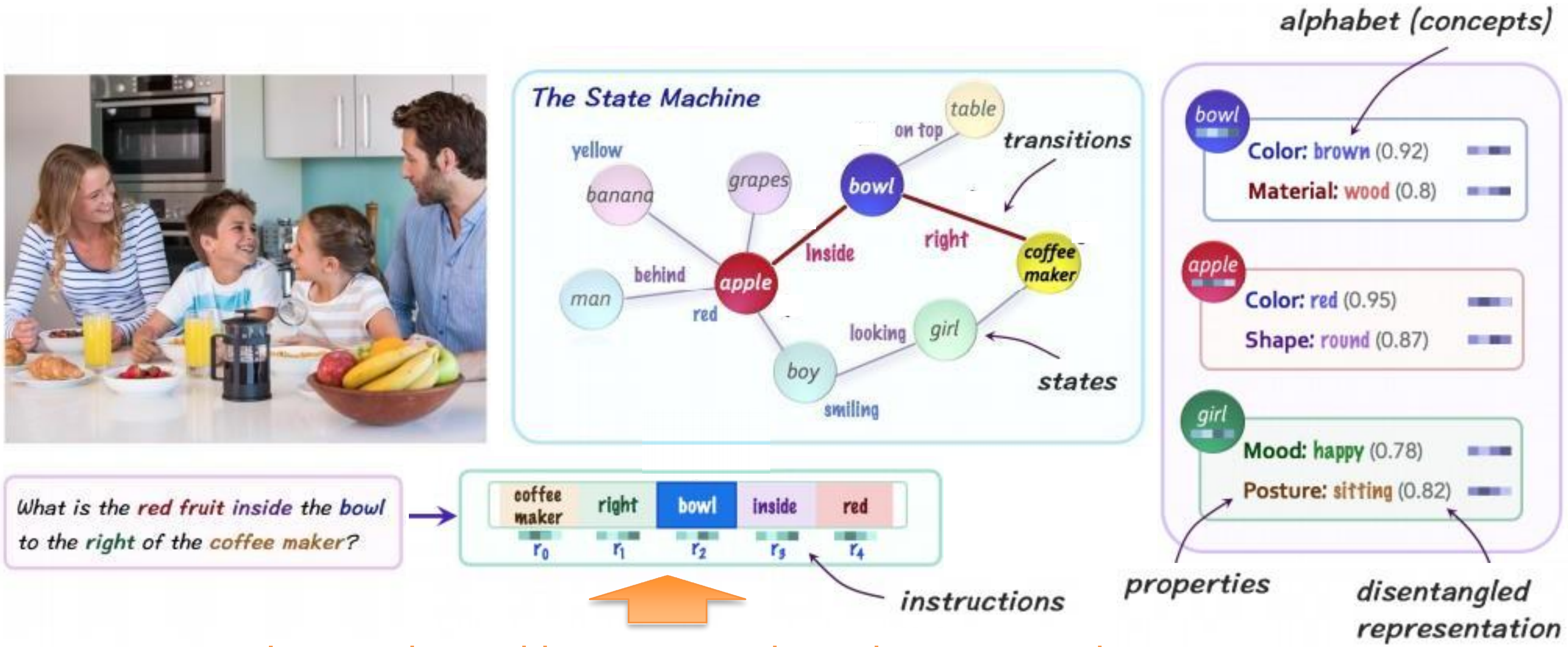
Module Networks V4: The Neural State Machine

Predefined an alphabet of relations
and compute probabilities for each directed edges



Hudson, Drew, and Christopher D. Manning. "Learning by abstraction: The neural state machine.“ NeurIPS 2019

Module Networks V4: The Neural State Machine

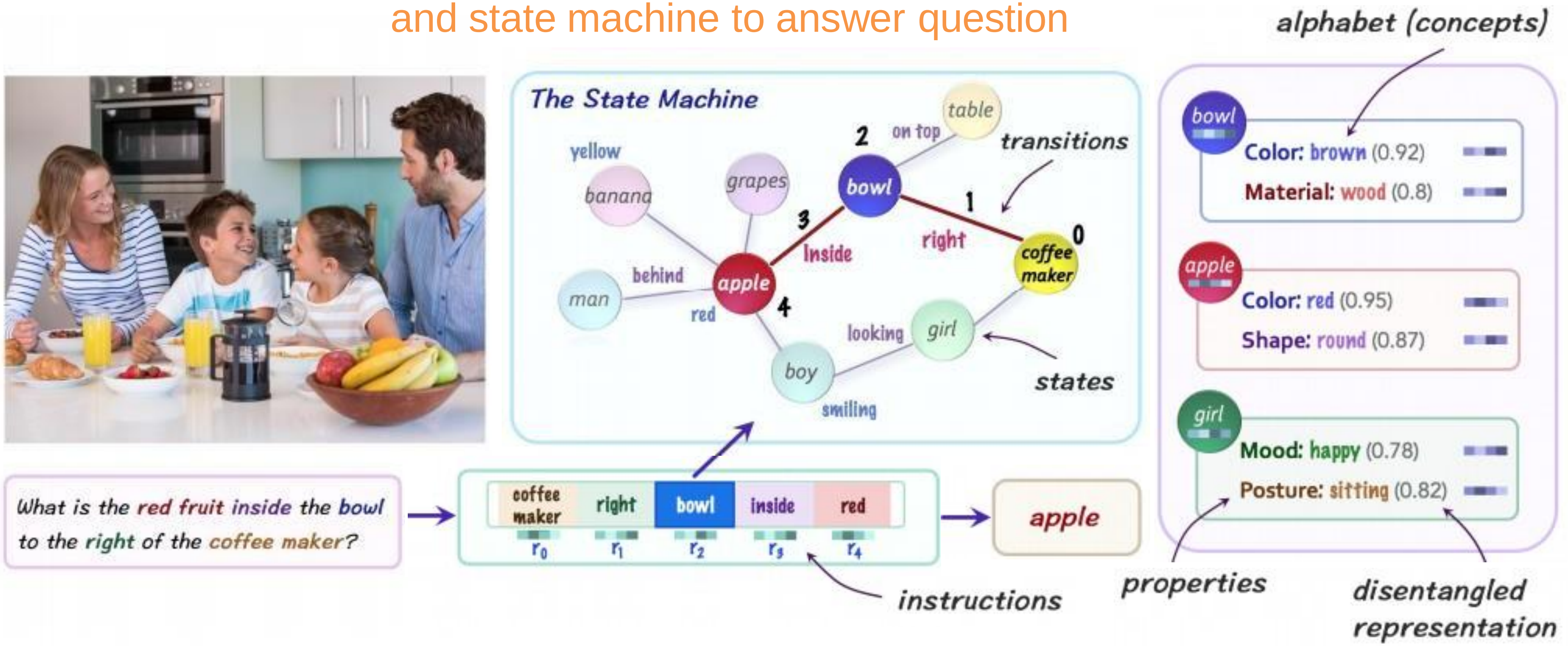


Translate each word in a concept-based representation
and group in a fixed number of instruction steps

Hudson, Drew, and Christopher D. Manning. "Learning by abstraction: The neural state machine." NeurIPS 2019

Module Networks V4: The Neural State Machine

Finally, perform reasoning using instructions
and state machine to answer question



Hudson, Drew, and Christopher D. Manning. "Learning by abstraction: The neural state machine.“ NeurIPS 2019